

Simultaneous Contact Selection and Planning for Contact-Rich Manipulation with Cascaded Optimization

Abstract—We propose an optimization-based framework for robust contact-rich manipulation. Recent contact-implicit methods enable online hybrid planning across contact modes, allowing closed-loop manipulation for a given target state and contact location sequence of the robot and object. However, most existing approaches lack the ability to autonomously reason and generate diverse contact location sequences and manipulation trajectories, i.e., active contact location selection, which limits their applicability to relatively simple tasks. Active contact location selection is challenging due to complementarity in contact dynamics and the sparse gradients, making the design of a unified framework for contact selection and planning difficult. To address these challenges, we introduce Simultaneous Contact Selection and Planning (SCSP), a cascaded optimization framework comprising Contact Selection Optimization (CSO) and Contact Planning Optimization (CPO). CSO leverages a surrogate contact model and discrete-continuous optimization to efficiently resolve the nonsmoothness and coupling in contact selection, enabling online **search over sampled contact locations**. CPO performs prior-guided contact planning by evaluating the reference contact locations produced by CSO and generating corresponding manipulation trajectories in real time for redundant manipulators. Extensive simulations and real-world experiments demonstrate that SCSP produces diverse manipulation behaviors and robust control under inaccurate dynamics and perceptual noise. We further validate the generalization of the framework on challenging manipulation tasks.

Project website: <https://sites.google.com/view/scsp-robot>.

Index Terms—Contact-Rich Manipulation, Contact Selection, Contact-Implicit Trajectory Optimization

NOMENCLATURE

$p, \mathbf{p}$	A 3D point and its position vector.
θ, ψ	Angle and quaternion vector.
$\mathcal{O}, \partial\mathcal{O}$	Object geometry and surface.
$\mathcal{R}, \partial\mathcal{R}$	Robot geometry and surface.
h	Time interval.
λ, β	Contact force vector and contact impulse vector.
ϕ	Contact distance vector.
$\mathbf{J}$	Contact Jacobian.
$\mathbf{n}, \mathbf{t}_i$	Normal vector and i -th tangential vector.
$\mathbf{x} = [\mathbf{p}, \psi]$	Pose vector consists of position and orientation.
$\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}}$	Robot configuration, velocity and acceleration.
$\mathbf{s}$	System state consists of object pose $\mathbf{x}_o$ and $\mathbf{q}$.
$\mathbf{v}$	Velocity vector in Cartesian space.

I. INTRODUCTION

CONTACT-Rich manipulation requires robots to intentionally make and break contact with objects to drive them toward desired states. Specifically, the robot needs to plan the

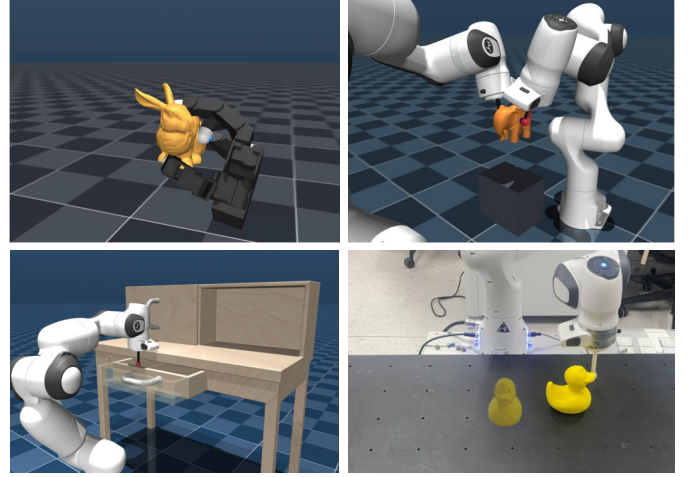


Fig. 1. SCSP enables online contact selection and closed-loop generation of diverse manipulation trajectories, allowing it to address a wide range of challenging contact-rich manipulation tasks.

contact sequence consisting of approaching trajectory, contact locations, contact forces, and transitions between different contact modes (i.e., contact, separation, sliding, and sticking). Such contact planning is challenging due to the combinatorial complexity of contact modes and the inherent nonsmoothness arising from the hybrid nature of contact dynamics. Early works adopted hierarchical frameworks by separating continuous-state planning and discrete contact mode searching [1]–[5], or utilizing mixed-integer programs to handle the hybrid nature of contact [6]–[8]. However, this separation often suffers from poor scalability caused by the explosion of contact trajectories. Therefore, these methods offer limited manipulation strategies and poor robustness.

Recent advances in contact-rich manipulation have been achieved through both learning-based and model-based methods. Learning-based methods [9]–[17] can learn complex contact-rich behaviors that are difficult to represent explicitly with model-based approaches. However, these methods typically rely on offline training, and their generalization is often demonstrated within specific task families, such as across unseen object geometries or variations of a predefined task. For example, DyWA [17] demonstrates impressive 6D non-prehensile manipulation across diverse object geometries, but its generalization is still evaluated within a specific manipulation setting. Consequently, achieving broadly applicable contact-rich manipulation across substantially different task settings remains challenging.

The representative model-based method is contact-implicit

TABLE I
COMPARISON OF CONTACT PLANNING METHODS.

Methods	Perception	Active Contact Selection	6D Manip. Exp.	Real-Time	No Obj.-Specific Analytic Geom.
C3 [18]	Exact State	✗	✗	✓	✗
CRISP [19]	Exact State	✗	✗	✓	✗
IDTO [20]	Exact State	✗	✓	✓	✗
CQDCs [21]	Exact State	✗	✓	✓	✗
CF-MPC [22]	Exact State	✗	✓	✓	✓
STOCS-3D [23]	Exact State	✗	✓	✗	✓
ContactSDF [24]	Vision	✗	✓	✓	✓
DyWA [17]	Vision	✗	✓	✓	✓
Sampling [25]	Vision	✓	✓	✓	✓
SCSP	Vision	✓	✓	✓	✓

trajectory optimization (CITO), which was first applied to locomotion [26], [27] and later extended to manipulation. STOCS-3D [23] extends CITO to high-fidelity geometries, but assumes a predefined manipulator contact state and is primarily designed for offline planning. We mainly focus on Contact-Implicit Model Predictive Control (CIMPC), as its receding-horizon formulation continuously replans from the current state and is therefore particularly suitable for real-time closed-loop contact-rich manipulation. CIMPC incorporates approximated time-stepping contact dynamics [28], [29] into trajectory optimization, enabling online hybrid contact planning. Representative methods improve the efficiency and scalability of this process through different contact-model approximations and optimization strategies [18], [21], [30]. In particular, CF-MPC [22] introduces a complementarity-free contact model to improve the efficiency of online contact-implicit MPC, substantially accelerating contact-rich optimization.

Despite these advances, existing contact-implicit planners mainly optimize how to execute locally available contact interactions, while robot–object contact locations are typically induced implicitly by robot controls and collision geometry or specified through contact priors. This limits their ability to actively explore spatially surface contacts from unfavorable contact-free configurations and makes them susceptible to poor local solutions [31], [32]. ContactSDF [24] develops differentiable multi-contact models for dexterous manipulation, but focuses on contact modeling rather than explicit robot–object contact-location selection. CRISP [19] can discover contact sequences within contact-implicit trajectory optimization, but is mainly demonstrated on relatively simple planar settings.

Only a few recent methods explicitly address robot–object contact-location exploration. Venkatesh et al. [25] sample candidate contact locations and evaluate the corresponding local CIMPC solutions, providing broader contact exploration at the cost of repeatedly solving the lower-level planner. Xie et al. [33] introduce a hierarchical RL–MPC framework in which a learned high-level policy selects contact locations for a low-level CIMPC controller, but the selector requires task-specific policy training.

Designing a unified planning framework for simultaneous contact selection and contact planning is challenging. The complementarity inherent in contact dynamics introduces non-

smoothness into contact selection. Moreover, since the choice of contact locations depends on the object state evolution at the time of contact, hybrid planning across contact-free and contact-rich stages renders contact selection a sparse-gradient optimization problem. A further challenge arises when connecting high-level contact selection with low-level execution: a contact location preferred according to the high-level manipulation objective is not necessarily the most executable contact at the current robot configuration. Strictly tracking such a contact prior can induce unnecessary contact breaking and switching, and can amplify errors caused by approximate dynamics, perception, or robot kinematic constraints. Therefore, contact selection should provide guidance to the planner rather than a hard contact command.

In this work, we propose Simultaneous Contact Selection and Planning (SCSP), a cascaded optimization framework for complex contact-rich manipulation. SCSP separates contact selection from trajectory generation to improve the diversity of manipulation strategies while preserving closed-loop adaptability during execution. The framework comprises two modules: Contact Selection Optimization (CSO) for generating task-directed contact priors and Contact Planning Optimization (CPO) for generating approaching, contact, and contact-switching trajectories.

The key design of SCSP lies in the coupling between contact selection and executable contact planning. CSO efficiently evaluates a finite set of sampled surface candidates using a lightweight surrogate contact model and selects a promising contact as the task-directed prior. Rather than treating this prior as a hard command, a ranking strategy compares it with the nearest contact candidate and determines the reference contact used by CPO. CPO then performs prior-guided closed-loop planning, providing a flexible interface between high-level contact selection and low-level trajectory generation.

Table I summarizes the scope of SCSP relative to representative contact-planning methods. The contributions are:

- A cascaded optimization framework, SCSP, is presented for contact-rich manipulation, comprising Contact Selection Optimization (CSO) and Contact Planning Optimization (CPO). The framework explicitly separates task-directed contact-prior generation from executable trajectory planning, enabling active contact selection while preserving closed-loop adaptability during execution.

- A ranking-guided interface is introduced to couple high-level contact selection with low-level contact planning. Rather than treating the contact prior as a hard command, the ranking strategy compares it with the nearest contact candidate and selects the reference contact online, allowing CPO to exploit acceptable suboptimal contacts while avoiding unnecessary contact switching. A unified formulation further supports approaching, contacting, breaking contact, and switching between contacts.
- An efficient CSO method is developed for online contact-prior generation over sampled object-surface candidates. A lightweight surrogate contact model enables fast candidate evaluation without requiring object-specific analytical geometric parameterizations.
- Extensive simulation and real-world experiments validate the effectiveness and robustness of SCSP across diverse contact-rich tasks and irregular object geometries, including structured multi-contact and long-horizon case studies. Real-world experiments further demonstrate vision-based 6D nonprehensile manipulation without markers or motion capture, with one-time calibration of the contact-estimation model.

The rest of the paper is organized as follows. Section II reviews the preliminaries. The CSO and CPO are introduced in Sections III and IV, respectively. Section VI presents the experimental results, and Section VII evaluates structured extensions of the SCSP architecture to additional contact-rich manipulation settings. Finally, discussion and limitations are presented in Section VIII, and conclusions are drawn in Section IX.

II. PRELIMINARIES

We begin with a brief overview of the contact-dynamics and contact-planning formulations used throughout this work. Section II-A introduces time-stepping contact dynamics and the associated complementarity formulation. Section II-B introduces model-based contact planning and reviews CIMPC as a representative online formulation. Section II-C reviews the complementarity-free contact model that motivates the lightweight contact-model approximation used later in CSO.

A. Contact Dynamics

Contact dynamics provides a physical description of interactions among the robot, object, and environment. Here, we consider the commonly used Anitescu model [29], which is formulated as the following time-stepping equation:

$$\epsilon M_o v_o = h \tau_o + \sum_{i=1}^{n_c} J_{o,i}^\top \beta_i, \quad (1a)$$

$$h K_r (h v_r - u) = h \tau_r + \sum_{i=1}^{n_c} J_{r,i}^\top \beta_i. \quad (1b)$$

Here, β_i denotes the contact impulse over the time interval h . In the contact-force formulation below, we use λ_i to denote the corresponding contact force. Assuming that the contact force is approximately constant within one time step, they are related by $\beta_i \approx h \lambda_i$. ϵM_o , v_o , τ_o , and $J_{o,i}$ are the regularized

mass matrix, velocity change, non-contact force, and contact Jacobian of the object, respectively. K_r , v_r , τ_r , and $J_{r,i}$ are the robot stiffness matrix, velocity, non-contact torque, and contact Jacobian, respectively. The control input is defined as $u = \Delta q$.

The above equations describe, from the perspective of momentum balance, how the robot influences the object's state through contact, as well as how the object interacts with the environment. Here, we follow [22], [34] and consider the robot to be under impedance control [35], modeling the robot-object interaction as a spring system. The contact constraints between contact forces and system motion are typically formulated as the following cone complementarity problem [36]:

$$\mathcal{K}_i \ni \lambda_i \perp J_i v + \frac{1}{h} \begin{bmatrix} \phi_i \\ 0 \\ 0 \end{bmatrix} \in \mathcal{K}_i^*, \quad \forall i \in \{1, \dots, n_c\}, \quad (2)$$

where $\mathcal{K}_i = \{\lambda_i \in \mathbb{R}^3 \mid \mu_i \lambda_i^n \geq \|\lambda_i^d\|\}$ is the Coulomb friction cone, and $\mathcal{K}_i^*$ is the dual cone as $\mathcal{K}_i^* = \{v \mid J_i^n v + \frac{\phi_i}{h} \geq \mu_i \|J_i^d v\|\}$.

Equations (1) and (2) are the Karush–Kuhn–Tucker (KKT) [37] optimality conditions of the following primal optimization [29]:

$$\min_v \quad \frac{1}{2} h^2 v^\top Q v - h v^\top b(u), \quad (3a)$$

$$\text{s.t.} \quad J_i v + \frac{1}{h} \begin{bmatrix} \phi_i \\ 0 \\ 0 \end{bmatrix} \in \mathcal{K}_i^*, \quad i \in \{1, \dots, n_c\}. \quad (3b)$$

where $Q \in \mathbb{R}^{(n_o + n_{\text{DoF}}) \times (n_o + n_{\text{DoF}})}$ and $b(u) \in \mathbb{R}^{n_o + n_{\text{DoF}}}$ are

$$Q := \begin{bmatrix} \epsilon M_o / h^2 & 0 \\ 0 & K_r \end{bmatrix}, \quad b(u) := \begin{bmatrix} \tau_o \\ K_r u + \tau_r \end{bmatrix}. \quad (4)$$

Here, n_o and n_{DoF} are the dimensions of the object and robot states, respectively, and n_c is the number of contacts.

Problem (3) is a second-order cone program (SOCP) and is often approximated by a QP through a polyhedral approximation of the friction cone:

$$(J_i^n - \mu_i J_{i,j}^d) v + \frac{\phi_i}{h} \geq 0, \quad i \in \{1, \dots, n_c\}, \quad j \in \{1, \dots, n_d\}, \quad (5)$$

where n_d is the number of friction-cone facets. Solving (3) corresponds to one step of physically consistent contact simulation. The resulting velocity v can then be used to integrate the current state as $q_{t+1} = q_t \oplus h v$. Problem (3) and its variants are widely used in physics simulators.

B. Model-based Contact Planning

Contact planning refers to generating trajectories for approaching, making contact, and breaking contact with the object to manipulate it to a target state. Model-based contact planning methods integrate contact dynamics and solve trajectory optimization problems. As a representative contact

planning method, CIMPC is formulated as the following optimization problem:

$$\begin{aligned} \min_{\mathbf{u}_{0:N-1}} \quad & \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + V_f(\mathbf{x}_N) \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, N-1, \\ & \mathbf{x}_k \in \mathcal{X}, \quad k = 0, \dots, N, \\ & \mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, N-1. \end{aligned} \quad (6)$$

in which $\mathbf{x}_k$ is the system state, including the object pose q_k and robot configuration, and f denotes the contact dynamics. The stage cost ℓ can take various forms and is typically used to guide the process of approaching the contact location. The terminal cost V_f is usually defined as the error between the object pose and the desired pose:

$$\ell_{\text{pose}} = w_{\text{pos}} \|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{goal}}\|_2^2 + w_{\text{quat}} \left(1 - (\boldsymbol{\psi}_{\text{obj}}^\top \boldsymbol{\psi}_{\text{goal}})^2\right). \quad (7)$$

The hybrid nature of contact dynamics renders (6) a piecewise-smooth nonlinear complementarity problem (NCP). Directly using the full contact model in (3) within receding-horizon trajectory optimization remains computationally demanding, motivating the efficient contact-model approximations introduced below.

C. Complementarity-free Model

We next introduce the complementarity-free contact model proposed in [22], which motivates the surrogate contact-model approximation used later in CSO. Problem (3) has the dual form:

$$\begin{aligned} \max_{\boldsymbol{\beta} \geq \mathbf{0}} \quad & -\frac{1}{2h^2} \boldsymbol{\beta}^\top (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\tilde{\mathbf{J}}^\top + \mathbf{R}) \boldsymbol{\beta} \\ & -\frac{1}{h} (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\mathbf{b} + \tilde{\boldsymbol{\phi}})^\top \boldsymbol{\beta} - \frac{1}{2} \mathbf{b}^\top \mathbf{Q}^{-1} \mathbf{b}. \end{aligned} \quad (8)$$

in which $\mathbf{R}$ is a small regularization term commonly used to enhance numerical stability, and

$$\tilde{\mathbf{J}} = \begin{bmatrix} \mathbf{J}_1^n - \mu_1 \mathbf{J}_{1,1}^t \\ \vdots \\ \mathbf{J}_1^n - \mu_1 \mathbf{J}_{1,n_t}^t \\ \vdots \\ \mathbf{J}_{n_c}^n - \mu_{n_c} \mathbf{J}_{n_c,1}^t \\ \vdots \\ \mathbf{J}_{n_c}^n - \mu_{n_c} \mathbf{J}_{n_c,n_t}^t \end{bmatrix}, \quad \tilde{\boldsymbol{\phi}} = \begin{bmatrix} \phi_1 \\ \vdots \\ \phi_1 \\ \vdots \\ \phi_{n_c} \\ \vdots \\ \phi_{n_c} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_{1,1} \\ \vdots \\ \beta_{1,n_d} \\ \vdots \\ \beta_{n_c,1} \\ \vdots \\ \beta_{n_c,n_d} \end{bmatrix}. \quad (9)$$

Here, $\boldsymbol{\beta}$ is not only a dual variable but also has a physical interpretation as the contact impulse. Specifically, the solution of (8) satisfies the following complementarity constraints:

$$\mathbf{0} \leq \boldsymbol{\beta} \perp \frac{1}{h} (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\tilde{\mathbf{J}}^\top + \mathbf{R}) \boldsymbol{\beta} + (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\mathbf{b} + \tilde{\boldsymbol{\phi}}) \geq \mathbf{0}. \quad (10)$$

Assuming there exists a positive-definite diagonal matrix that satisfies $\mathbf{K}(\mathbf{q}) = (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\tilde{\mathbf{J}}^\top + \mathbf{R})^{-1}$, we get the closed-form solution of (8) as:

$$\boldsymbol{\beta}^+ = \max \left(-h\mathbf{K}(\mathbf{q}) (\tilde{\mathbf{J}}\mathbf{Q}^{-1}\mathbf{b} + \tilde{\boldsymbol{\phi}}), \mathbf{0} \right). \quad (11)$$

The complementarity-free model is then written as:

$$f_{\text{cf}} : \begin{cases} \mathbf{v}^+ = \frac{1}{h} \mathbf{Q}^{-1} \mathbf{b} + \frac{1}{h} \mathbf{Q}^{-1} \tilde{\mathbf{J}}^\top \boldsymbol{\beta}^+, \\ \mathbf{x}_{k+1} = \mathbf{x}_k + h(\mathbf{v}_k + \mathbf{v}^+). \end{cases} \quad (12)$$

where $\boldsymbol{\beta}$ denotes the contact impulses of all contact points and $\mathbf{v}^+$ combines the object velocity $\mathbf{v}_o$ and robot velocity $\mathbf{v}_r$.

As discussed in [22], this assumption provides a computationally efficient approximation of the original contact model and enables the closed-form update in (11).

III. CONTACT SELECTION OPTIMIZATION

In this section, we present the detailed design of CSO. We begin by analyzing the key challenges of integrating contact selection into CIMPC, which limit the diversity of existing methods. We then formulate the contact selection problem and discuss the main difficulties in solving it. Finally, we present the detailed design of our CSO module for online contact selection.

A. Why Contact Selection is necessary?

Contact selection refers to choosing the contact locations $\{\mathbf{p}_{\mathcal{O},i}\} \in \partial\mathcal{O}$ and $\{\mathbf{p}_{\mathcal{R},i}\} \in \partial\mathcal{R}$ according to a given task specification. Since $\partial\mathcal{O}$ is a low-dimensional manifold embedded in Cartesian space, the set of robot-object contact states (i.e., the contact-rich stage) also lies on a low-dimensional manifold $\mathcal{S}_{\text{contact}}$ within the full state space $\mathcal{S}$. Task-related costs are defined only on $\mathcal{S}_{\text{contact}}$, whereas the remaining state space $\mathcal{S}_{\text{free}}$ is gradient-sparse. Moreover, CIMPC methods typically employ a short¹ and fixed planning horizon with a fixed step size, which prevents reliable exploration of cost differences arising from different contact locations when starting from an arbitrary contact-free state $\mathbf{s}_k \in \mathcal{S}_{\text{free}}$. As a result, integrating contact selection into CIMPC, which must hybridly plan both contact-free and contact-rich trajectories, is challenging. Existing CIMPC methods often rely on manually designed costs or reference trajectories to guide contacts. Such designs effectively fix the contact locations and constrain the resulting manipulation trajectories to a narrow distribution, limiting the diversity of achievable behaviors. Moreover, the contact location is not a direct control variable in robot-centric CIMPC but is determined implicitly through nonsmooth collision checking, which further introduces complexity.

Rather than integrating contact selection into the CIMPC formulation, we study how to formulate contact selection as a separate optimization problem. As illustrated in Fig. 2, we first simplify the problem by ignoring the planning of the robot contact locations $\{\mathbf{p}_{\mathcal{R},i}\}$ and thereby obtain the following object-centric optimization problem:

$$\begin{aligned} \min_{\mathbf{p}_r \in \partial\mathcal{O}, \boldsymbol{\lambda} \in \mathcal{K}} \quad & \ell_{\text{cso}}(\mathbf{x}_T) \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{p}_r, \boldsymbol{\lambda}), \quad k = 0, \dots, T-1. \end{aligned} \quad (13)$$

where $\mathcal{K}$ is the set of feasible contact forces and $\ell_{\text{cso}}(\cdot)$ is the objective of contact selection. $\mathbf{p}_r \in \mathbb{R}^{3 \times n_r}$ represents

¹For computational efficiency, these methods usually adopt a local approximation of the contact dynamics, which limits the valid region (i.e., the trust region) of the model and therefore necessitates a short planning horizon.

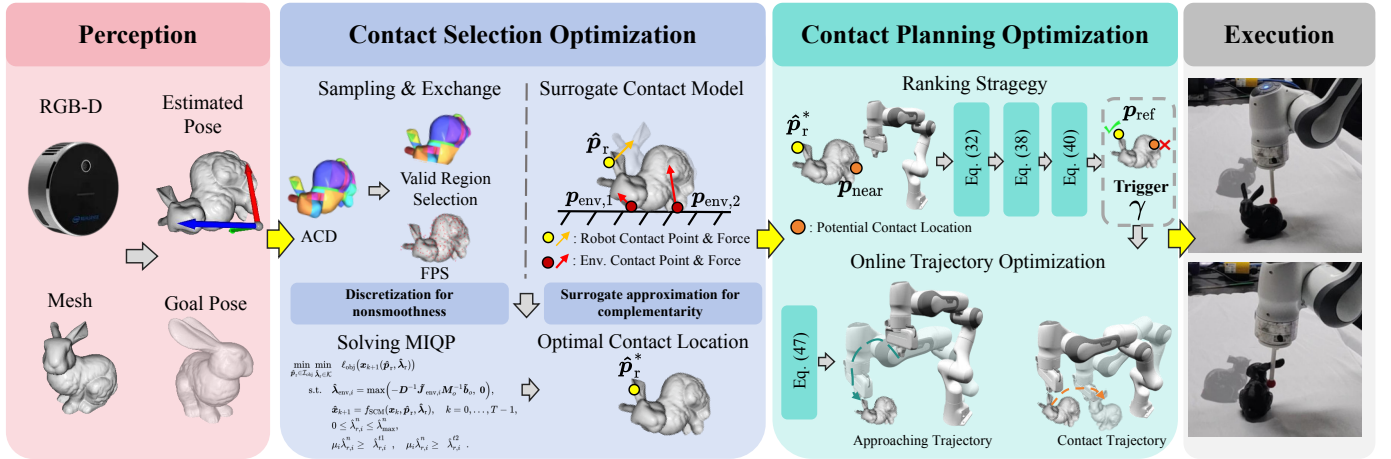


Fig. 2. Framework structure of SCSP. SCSP consists of Contact Selection Optimization (CSO) and Contact Planning Optimization (CPO). CSO discretizes nonsmooth geometry through the Sampling & Exchange method and introduces a surrogate contact model for efficient approximation of contact dynamics. CSO online computes the **optimal contact location among candidate contact set** $\hat{\mathbf{p}}_r^*$ as contact prior. The ranking strategy in CPO further evaluates whether the nearest point $\mathbf{p}_{\text{near}}$ can serve as a feasible suboptimal substitute for $\hat{\mathbf{p}}_r^*$ as the reference contact location, after which trajectory optimization is solved online to generate the approach and contact trajectories.

the locations where the robot applies contacts to the object, with n_r denotes the number of robot contacts. (13) addresses the gradient sparsity issue arising from hybrid planning in the original contact-rich manipulation problem by defining CSO to solve for **low-cost sampled contact locations** and contact forces on $\mathcal{S}_{\text{contact}}$. The neglected process of the robot approaching $\mathcal{S}_{\text{contact}}$ from its current state is the primary source of the gradient sparsity. Note that (13) is hard to solve due to the coupling of contact location $\mathbf{p}_r$ and the contact force $\boldsymbol{\lambda} = [\boldsymbol{\lambda}_{\text{env}}, \boldsymbol{\lambda}_r]$. Therefore, we adopt a commonly used simplification for handling nonconvex optimization problems to reformulate (13) as subproblems:

$$\min_{\mathbf{p}_r \in \partial \mathcal{O}} V(\mathbf{p}_r), \quad (14)$$

in which:

$$\begin{aligned} V(\mathbf{p}_r) := & \min_{\boldsymbol{\lambda}_r \in \mathcal{K}} \ell_{\text{cso}}(\mathbf{x}_T) \\ \text{s.t. } & \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{p}_r, \boldsymbol{\lambda}_r, \boldsymbol{\lambda}_{\text{env}}), \quad k = 0, \dots, T-1 \end{aligned} \quad (15)$$

The physical interpretation of (14) and (15) is that a contact location is regarded as optimal if applying the best possible combination of contact force $\boldsymbol{\lambda}^*$ yields the minimum motion cost of the object. Despite the simplifications, (14) remains a complex problem with nonsmooth constraints $\mathbf{p}_r \in \partial \mathcal{O}$ that depend on the object geometry. Meanwhile, (15) is a complementary problem with coupled environment forces $\boldsymbol{\lambda}_{\text{env}}$ and robot contact forces $\boldsymbol{\lambda}_r$. The nonconvexity makes online CSO intractable. We next present the solutions to further simplify both (14) and (15) to improve computational efficiency.

B. Sampling and Exchange Method

The nonsmoothness on the object geometry leads to discontinuities in the gradient of (14). The inherently nonsmooth nature of the problem motivates us to adopt a sampling-based method. To simplify (14), we uniformly sample a set

of candidate contact point set $\mathcal{I}_{\text{obj}} = \{\mathbf{p}_i\}_{i=1}^{n_s}$ on the object surface $\partial \mathcal{O}$, thus reformulate (14) as:

$$\min_{\mathbf{p}_r \in \mathcal{I}_{\text{obj}}} V(\mathbf{p}_r), \quad (16)$$

When the sampling is sufficiently uniform, (16) provides a good approximation to (14) and is not affected by discontinuous gradients caused by the geometry. To obtain a high quality candidate set, $\mathcal{I}_{\text{obj}}$ should satisfy the following conditions:

- **Physical reachability**, which requires that candidate contact locations be reachable by the robot, i.e., within the robot workspace and not occluded.
- **Contact stability**, which requires avoiding regions with high curvature where stable contact is difficult to apply.
- **Uniform distribution**, which means the sampled points are representative of approximating the original geometry.

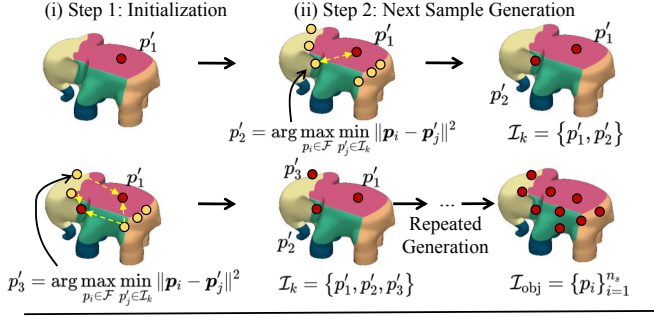
These conditions require the sampling method to maintain uniform coverage while prioritizing manipulability over local geometric details. To address this, we propose an efficient sampling strategy as shown in Algorithm 1. We first perform approximate convex decomposition (ACD) [38] to obtain a convex approximation $\mathcal{F}$ of the original geometry $\mathcal{O}$. ACD decomposes the object into convex polygons, mitigating interference caused by local geometric details and reducing computational cost. Moreover, these polygons are typically larger and flatter, which implies greater contact stability and improved reachability.

Then we perform Farthest-Point Sampling (FPS) on the convex approximation to uniformly sample points on the polygons. Starting from an arbitrary point as a sample, the algorithm iteratively selects the points that maximize the minimal Euclidean distance to all previously chosen samples:

$$\mathbf{p}'_{k+1} = \arg \max_{\mathbf{p}_i \in \mathcal{F}} \min_{\mathbf{p}'_j \in \mathcal{I}_k} \|\mathbf{p}_i - \mathbf{p}'_j\|^2, \quad (17)$$

where $\mathcal{I}_k$ is the set of already selected samples. The sampling process is performed offline. Solving (17) yields a uniformly distributed set of points $\mathcal{I}_{\text{obj}}$. For each sampled point $\mathbf{p}'_j \in$

(a) Farthest-Point Sampling (Offline):



(b) Valid Region Selection (Online):

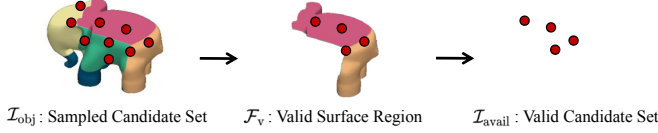


Fig. 3. The sampling strategy in CSO. The candidate point set is sampled sequentially through: (a) Farthest-Point Sampling to uniformly sample points on the object, and (b) Valid Region Selection to select valid points for contact.

$\mathcal{I}_k$, we also construct its local coordinate frame to obtain the tangential and normal vectors as local contact information. The local orthonormal frame is constructed as $(\mathbf{n}_i, \mathbf{t}_{1,i}, \mathbf{t}_{2,i})$ in which the normal vector $\mathbf{n}_i$ is obtained from the mesh or point-cloud normal estimate, and two orthogonal tangent directions are computed as:

$$\mathbf{t}_{1,i} = \frac{\mathbf{a} - (\mathbf{a} \cdot \mathbf{n}_i)\mathbf{n}_i}{\|\mathbf{a} - (\mathbf{a} \cdot \mathbf{n}_i)\mathbf{n}_i\|}, \quad \mathbf{t}_{2,i} = \mathbf{n}_i \times \mathbf{t}_{1,i}, \quad (18)$$

where $\mathbf{a}$ is an arbitrary fixed axis. **When the reference axis $\mathbf{a}$ is nearly parallel to the surface normal $\mathbf{n}_i$, we select another coordinate axis as the reference axis to avoid numerical singularity.** We use this local geometric information to compute the Jacobian required for contact force optimization. Furthermore, a KD-tree $\mathcal{T} = \{(\mathbf{p}_i, \mathbf{n}_i, \mathbf{t}_{1,i}, \mathbf{t}_{2,i})\}_{i=1}^{n_s}$ is constructed over the set $\mathcal{I}_{\text{obj}}$ to enable efficient nearest-neighbor queries.

An online Valid Region Selection procedure is further applied to identify feasible and task-relevant surface regions, filtering out unreachable or low-quality points such as occluded regions, regions outside the robot workspace, and **regions that violate task-specific semantic masks (for example, handles or forbidden surfaces)**. The valid region is maintained by applying a mask to $\mathcal{F}$ to obtain $\mathcal{F}_v$, and then selecting points on $\mathcal{F}_v$ to obtain $\mathcal{I}_{\text{avail}}$. To better illustrate the process, consider the toy example shown in Fig. 3. First, following the procedure illustrated in Fig. 3(a), we iteratively apply (17) to obtain a set of uniformly distributed sampling points over the processed object mesh. We then partition the surface online into a set of valid regions and masked regions, and select the points that lie within the valid regions to form the candidate set $\mathcal{I}_{\text{avail}}$, as shown in Fig. 3(b).

C. Surrogate Contact Model

The sampling strategy transforms the original optimization into a discrete-continuous optimization problem. We further simplify (15) to improve efficiency. Most existing

Algorithm 1 Sampling and Exchange Method

Input: Object geometry $\mathcal{O}$, sample count K , fixed axis $\mathbf{a}$
Output: $\mathcal{T} = \{(\mathbf{p}_i, \mathbf{n}_i, \mathbf{t}_{1,i}, \mathbf{t}_{2,i})\}_{i=1}^{n_s}$ and $\mathcal{I}_{\text{avail}}$.

- 1: $\mathcal{F} \leftarrow \text{ACD}(\mathcal{O})$ ▷ Convex approximation
- 2: Initialize $\mathcal{I}_k \leftarrow \{p'_1\}$ with an arbitrary seed in $\mathcal{F}$
- 3: **for** $k = 1$ to $K - 1$ **do**
- 4: $p'_{k+1} \leftarrow \arg \max_{p'_i \in \mathcal{F}} \min_{p'_j \in \mathcal{I}_k} \|\mathbf{p}'_i - \mathbf{p}'_j\|^2$
- 5: $\mathcal{I}_{\text{obj}} \leftarrow \mathcal{I}_k \cup \{p'_{k+1}\}$ ▷ Obtain candidate set $\mathcal{I}_{\text{obj}}$
- 6: **end for**
- 7: $\mathcal{F}_v \leftarrow \text{getValidRegion}(\mathcal{F})$ ▷ Task-specific segment
- 8: **for all** $p_i \in \mathcal{I}_{\text{obj}}$ **do**
- 9: **if** $p_i \notin \mathcal{F}_v$ **then**
- 10: Remove p_i from $\mathcal{I}_{\text{obj}}$ ▷ Valid region selection
- 11: **continue**
- 12: **end if** ▷ Obtain $\mathcal{I}_{\text{avail}}$
- 13: Estimate normal $\mathbf{n}_i$ from $\mathcal{F}$ at p_i
- 14: $\mathbf{t}_{1,i} \leftarrow \frac{\mathbf{a} - (\mathbf{a} \cdot \mathbf{n}_i)\mathbf{n}_i}{\|\mathbf{a} - (\mathbf{a} \cdot \mathbf{n}_i)\mathbf{n}_i\|}, \quad \mathbf{t}_{2,i} \leftarrow \mathbf{n}_i \times \mathbf{t}_{1,i}$
- 15: **end for**
- 16: **return** $\mathcal{T} = \{(\mathbf{p}_i, \mathbf{n}_i, \mathbf{t}_{1,i}, \mathbf{t}_{2,i})\}_{i=1}^{n_s}$

contact models are computationally expensive for scenarios that require repeatedly solving (15). Moreover, the complementarity in contact dynamics and the coupling between λ_{env} and λ_r make the optimization difficult to converge, leading to physically inconsistent results. To address this issue and enable online contact selection, we present a surrogate contact model (SCM) as a local approximation of the original contact dynamics. The procedure of SCM is illustrated in Fig. 4. The approximation relies on the following assumptions of the object-environment contact:

Assumption 1 (Fixed Contact Set). The object-environment contact location remains unchanged in each SCM rollout, i.e., no new contacts are created and existing contacts remain.

Assumption 2 (Multi-Contact Decoupling). The contact between the object and the environment can be approximated by independent spring-damper systems, neglecting higher-order dynamic couplings between different contact locations.

We reformulate the original contact dynamics in (1a) as:

$$\epsilon M_o \mathbf{v}_o = h(\tau_o + \sum_{i=1}^{n_r} \mathbf{J}_{r,i}^\top \lambda_{r,i} + \sum_{j=1}^{n_{\text{env}}} \mathbf{J}_{\text{env},j}^\top \lambda_{\text{env},j}), \quad (19)$$

and we denote $\tilde{\mathbf{b}}_o = h(\tau_o + \sum_{i=1}^{n_r} \mathbf{J}_{r,i}^\top \lambda_{r,i})$. Similar to (8), the physically consistent environment contact force can be obtained by solving the following dual problem:

$$\begin{aligned} \max_{\lambda_{\text{env}} \geq 0} \quad & -\frac{1}{2} \lambda_{\text{env}}^\top \left(\tilde{\mathbf{J}}_{\text{env}} M_o^{-1} \tilde{\mathbf{J}}_{\text{env}}^\top + \mathbf{R} \right) \lambda_{\text{env}} \\ & - (\tilde{\mathbf{J}}_{\text{env}} M_o^{-1} \tilde{\mathbf{b}}_o + \tilde{\phi})^\top \lambda_{\text{env}} - \frac{1}{2} \tilde{\mathbf{b}}_o^\top M_o^{-1} \tilde{\mathbf{b}}_o, \end{aligned} \quad (20)$$

and the solution satisfies the complementarity condition

$$\mathbf{0} \leq \lambda_{\text{env}} \perp \mathbf{W}_{\text{env}} \lambda_{\text{env}} + \mathbf{g}_{\text{env}} \geq \mathbf{0}. \quad (21)$$

where

$$\mathbf{W}_{\text{env}} = \left(\tilde{\mathbf{J}}_{\text{env}} M_o^{-1} \tilde{\mathbf{J}}_{\text{env}}^\top + \mathbf{R} \right), \quad \mathbf{g}_{\text{env}} = \left(\tilde{\mathbf{J}}_{\text{env}} M_o^{-1} \tilde{\mathbf{b}}_o + \tilde{\phi} \right). \quad (22)$$

Here $\mathbf{W}_{\text{env}}$ is the Delassus matrix associated with object–environment contact. Since the contact force at each contact point is decomposed into one normal and two tangential components $\lambda_i = [\lambda_i^n, \lambda_i^{t_1}, \lambda_i^{t_2}]^\top \in \mathbb{R}^3$, the Delassus matrix $\mathbf{W}_{\text{env}}$ admits a block structure

$$\mathbf{W}_{\text{env}} = \begin{bmatrix} \mathbf{W}_{11} & \mathbf{W}_{12} & \cdots & \mathbf{W}_{1n_{\text{env}}} \\ \mathbf{W}_{21} & \mathbf{W}_{22} & \cdots & \mathbf{W}_{2n_{\text{env}}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{W}_{n_{\text{env}}1} & \mathbf{W}_{n_{\text{env}}2} & \cdots & \mathbf{W}_{n_{\text{env}}n_{\text{env}}} \end{bmatrix},$$

where each block $\mathbf{W}_{ij} \in \mathbb{R}^{3 \times 3}$ is

$$\mathbf{W}_{ij} = \begin{bmatrix} \mathbf{J}_i^n \\ \mathbf{J}_i^{t_1} \\ \mathbf{J}_i^{t_2} \end{bmatrix} \mathbf{M}_o^{-1} [(\mathbf{J}_j^n)^\top \quad (\mathbf{J}_j^{t_1})^\top \quad (\mathbf{J}_j^{t_2})^\top] + \mathbf{R}_{ij}.$$

The diagonal blocks $\mathbf{W}_{ii}$ describe the local effective inertia at each contact, indicating how forces applied at contact i affect motion along its own constraint directions. The off-diagonal blocks $\mathbf{W}_{ij}$ ($i \neq j$) represent dynamic coupling between different contacts. Under Assumption 2, these inter-contact couplings are negligible. Therefore, $\mathbf{W}_{\text{env}}$ can be approximated by a block-diagonal matrix:

$$\mathbf{W}_{\text{env}} = \text{blockdiag}(\mathbf{W}_{11}, \mathbf{W}_{22}, \dots, \mathbf{W}_{n_{\text{env}}n_{\text{env}}}), \quad (23)$$

and (21) is decomposed to subproblems:

$$\mathbf{0} \leq \lambda_{\text{env},i} \perp \mathbf{W}_{ii} \lambda_{\text{env},i} + (\tilde{\mathbf{J}}_{\text{env},i} \mathbf{M}_o^{-1} \tilde{\mathbf{b}}_o + \tilde{\phi}_i) \geq \mathbf{0}. \quad (24)$$

Furthermore, if $\mathbf{W}_{ii}$ admits the low-rank decomposition:

$$\mathbf{W}_{ii} = \mathbf{D} + \mathbf{E}, \quad (25)$$

where $\mathbf{D}$ is a diagonal matrix and $\mathbf{E}$ is relatively small, (24) has the approximated closed-form solution:

$$\hat{\lambda}_{\text{env},i} = \max \left(-\mathbf{D}^{-1} (\tilde{\mathbf{J}}_{\text{env},i} \mathbf{M}_o^{-1} \tilde{\mathbf{b}}_o + \tilde{\phi}_i), \mathbf{0} \right). \quad (26)$$

With (26), we can obtain the following lemma:

Lemma 1. Under Assumption 2, the environment contact force λ_{env} is an explicit function of λ_r .

To obtain a high-quality approximation, we decompose $\mathbf{W}_{ii}$ in (25) as:

$$\mathbf{D} = \text{diag}(\mathbf{W}_{ii}) + \epsilon \mathbf{I}, \quad (27)$$

where ϵ is a small regularization coefficient to ensure numerical stability. (27) gives the closest diagonal approximation to $\mathbf{W}_{ii}$ in the Frobenius norm². However, it should be noted that (26) is incompatible with gradient-based solvers. When the contact set changes within the CSO horizon, variations in $\tilde{\mathbf{J}}_{\text{env},i}$ and $\tilde{\phi}_i$ introduce additional non-smoothness. To address this, we leverage Assumption 1 to further derive:

Lemma 2. The environment contact force λ_{env} is a piecewise linear function of λ_r :

$$\hat{\lambda}_{\text{env},i} = \max \left(-\mathbf{D}^{-1} \tilde{\mathbf{J}}_{\text{env},i} \mathbf{M}_o^{-1} (\tau_o + \sum_{i=1}^{n_r} \mathbf{J}_{r,i}^\top \hat{\lambda}_{r,i}), \mathbf{0} \right). \quad (28)$$

²Proof see Appendix A.

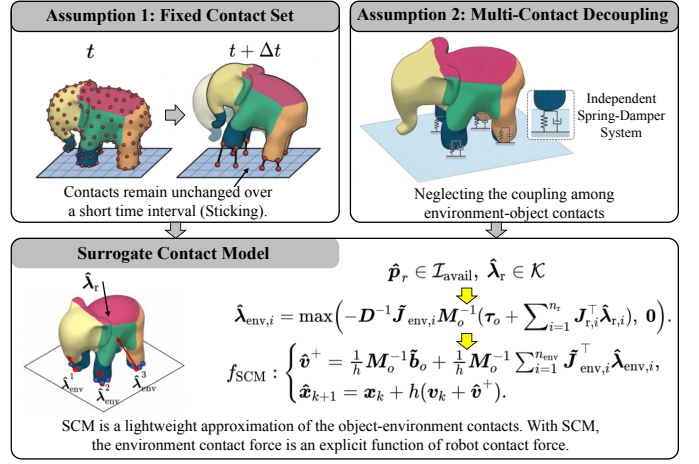


Fig. 4. Illustration of SCM. SCM is a lightweight approximation of the object-environment contact, enabling fast contact selection optimization.

Proof. Based on Assumption 1, we can obtain that $\phi \equiv 0$, leading to (28). Moreover, since the contact geometry set remains invariant over the time interval, $\tilde{\mathbf{J}}_{\text{env},i}$ is a constant matrix, thus $\mathbf{W}_{ii}$ is constant, and $\mathbf{D}$, $\mathbf{M}_o^{-1}$ are all constant matrices. Therefore, (28) is piecewise linear.

Substituting (28) into (19), we obtain a lightweight explicit model:

$$f_{\text{SCM}} : \begin{cases} \hat{\mathbf{v}}^+ = \frac{1}{h} \mathbf{M}_o^{-1} \tilde{\mathbf{b}}_o + \frac{1}{h} \mathbf{M}_o^{-1} \sum_{i=1}^{n_{\text{env}}} \tilde{\mathbf{J}}_{\text{env},i}^\top \hat{\lambda}_{\text{env},i}, \\ \hat{\mathbf{x}}_{k+1} = \mathbf{x}_k + h(\mathbf{v}_k + \hat{\mathbf{v}}^+). \end{cases} \quad (29)$$

This Surrogate Contact Model converts the original contact dynamics into a computationally efficient formulation, enabling (15) to be solved efficiently using gradient-based solvers.

Lemma 3 (Local prediction estimates for SCM). Consider the reference model (29) and SCM in (29), evaluated using identical current states, robot contact inputs, contact geometry, and model parameters in the zero-gap setting of (28). Let λ_{env} and $\hat{\lambda}_{\text{env}}$ denote their nonnegative contact multipliers. Assume $\mathbf{M}_o \succ 0$ and $R \succ 0$, let $\mathbf{D} \succ 0$ collect the diagonal approximations in (27), and define $\mathbf{E} := \mathbf{W}_{\text{env}} - \mathbf{D}$.

The predicted velocity changes satisfy

$$(\mathbf{v}^+)^\top \mathbf{M}_o \hat{\mathbf{v}}^+ \geq \|\mathbf{v}^+\|_{\mathbf{M}_o}^2 - \frac{\hat{\lambda}_{\text{env}}^\top R \hat{\lambda}_{\text{env}}}{4h^2}.$$

Their multiplier and motion discrepancies satisfy

$$\begin{aligned} \|\hat{\lambda}_{\text{env}} - \lambda_{\text{env}}\|_{\mathbf{D}} &\leq \|\mathbf{D}^{-1/2} \mathbf{E} \lambda_{\text{env}}\|_2, \\ h\|\hat{\mathbf{v}}^+ - \mathbf{v}^+\|_{\mathbf{M}_o} &\leq \Gamma_{\text{env}} \|\mathbf{D}^{-1/2} \mathbf{E} \lambda_{\text{env}}\|_2, \end{aligned}$$

where $\|z\|_A = (z^\top A z)^{1/2}$ and

$$\Gamma_{\text{env}} := \|\mathbf{M}_o^{-1/2} \tilde{\mathbf{J}}_{\text{env}}^\top \mathbf{D}^{-1/2}\|_2.$$

Proof. See Appendix B. □

D. Formulation of CSO

Integrating SCM into (15), we can obtain the formulation of CSO as:

$$\begin{aligned}
& \min_{\hat{\mathbf{p}}_r \in \mathcal{I}_{\text{avail}}} \min_{\hat{\lambda}_r \in \mathcal{K}} \ell_{\text{CSO}}(\mathbf{x}_T) \\
& \text{s.t. } \hat{\lambda}_{\text{env},i} = \max \left(-D^{-1} \tilde{\mathbf{J}}_{\text{env},i} M_o^{-1} \tilde{\mathbf{b}}_o, \mathbf{0} \right), \\
& \hat{\mathbf{x}}_{k+1} = f_{\text{SCM}}(\mathbf{x}_k, \hat{\mathbf{p}}_r, \hat{\lambda}_r), \quad k = 0, \dots, T-1, \\
& |\hat{\lambda}_{r,i}^{t1}| \leq \mu_i \hat{\lambda}_{r,i}^n, \quad |\hat{\lambda}_{r,i}^{t2}| \leq \mu_i \hat{\lambda}_{r,i}^n, \\
& 0 \leq \hat{\lambda}_{r,i}^n \leq \hat{\lambda}_{\text{max}}^n, \quad i = 0, \dots, n_r.
\end{aligned} \tag{30}$$

By solving this discrete-continuous optimization problem, we can obtain the **optimal contact location among candidate contact set** $\hat{\mathbf{p}}_r^* \in \mathcal{I}_{\text{avail}}$ as guidance for the lower-level contact planning. Next, we discuss the details of CSO.

1) *Objective*: Despite the objective of CSO being to minimize the object pose cost as denoted in (7), directly using ℓ_{pose} as the CSO objective ℓ_{CSO} results in unstable contact strategies by blindly minimizing cost. In addition, the execution of the lower-level contact planning approach is inherently subject to delay, since the robot must physically approach and establish contact rather than instantaneously reaching the selected contact location. Therefore, the design of the CSO should explicitly consider the delay and avoid unstable behaviors that may cause frequent dynamic changes in the object pose. Therefore, we introduce the concept of an equivalence class of stable poses of an object as $\mathcal{X}_{\text{stab}} = \bigcup_{j=1}^k \mathcal{X}_j$, where $\mathcal{X}_j$ denotes the j -th class of stable pose of the object and $\mathbf{x}_o \in \mathcal{X}_j$ is a statically stable pose at which the object can remain at rest on the table, with its center of mass projected within the support polygon. We define the transfer cost from the current object pose to the goal stable pose as:

$$\ell_{\text{stab}}(\mathbf{x}_o) = d(\mathbf{x}_o, \mathcal{X}_g), \tag{31}$$

where $d(\cdot, \cdot)$ is a pose distance metric, $\mathcal{X}_j$ denotes the set of object poses associated with the j -th statically stable placement mode, and $\alpha > 0$ is a weighting coefficient. We reformulate the CSO objective to prioritize reaching the goal pose set $\mathcal{X}_g$ from the current pose $\mathbf{x}_o$, and then minimize ℓ_{pose} in $\mathcal{X}_g$:

$$\ell_{\text{CSO}} = \begin{cases} \ell_{\text{stab}}, & \text{if } \mathbf{x}_o \notin \mathcal{X}_g, \\ \ell_{\text{pose}}, & \text{otherwise.} \end{cases} \tag{32}$$

The design of (32) effectively guides the gradient descent of CSO to select a more statically stable trajectory. Although this increases the control cost, it effectively prevents CSO from generating numerous unstable contact locations as priors, which would otherwise make the lower-level contact planning difficult to execute in practice.

2) *Solving CSO over Sampled Candidates*: The original contact dynamics in (15) are non-differentiable because of the complementarity constraints and the non-smooth temporal evolution of the contact set, rendering the constraints non-differentiable. With the SCM, (30) becomes a finite candidate

search with a continuous force-optimization subproblem for each candidate. The SCM gradients can be computed via:

$$\begin{aligned}
\frac{\partial \mathbf{x}_{k+1}}{\partial \lambda_{r,i}} &= \frac{\partial \mathbf{x}_{k+1}}{\partial \mathbf{v}^+} \left(\frac{\partial \mathbf{v}^+}{\partial \mathbf{b}} \frac{\partial \mathbf{b}}{\partial \lambda_r} + \frac{\partial \mathbf{v}^+}{\partial \lambda_{\text{env}}} \frac{\partial \lambda_{\text{env}}}{\partial \lambda_r} \right) \\
&= M_o^{-1} \left(h \mathbf{J}_{r,i}^\top + \frac{1}{h} \sum_{j=0}^{n_{\text{env}}} \tilde{\mathbf{J}}_{\text{env},j}^\top \frac{\partial \lambda_{\text{env},j}}{\partial \lambda_{r,i}} \right),
\end{aligned} \tag{33}$$

where the gradient $\frac{\partial \lambda_{\text{env},j}}{\partial \lambda_{r,i}}$ can be computed via automatic differentiation and satisfies Lipschitz continuity because (28) is piecewise linear. Moreover, the objective function ℓ_{CSO} is second-order differentiable. Therefore, the continuous subproblem can be solved efficiently for each candidate; the reported implementation uses SNOPT to solve the resulting nonlinear program, and the candidate costs are then enumerated.

IV. CONTACT PLANNING OPTIMIZATION

In this section, we introduce the CPO module for contact planning. CPO accepts the optimal contact location predicted by CSO as reference, and generates approaching and contact trajectories online. We will discuss the problem of designing a prior-guided contact planning. Then we design a ranking strategy as an expressive connection between contact planning and contact prior from contact selection module. Finally, we propose a novel objective formulation for prior-guided contact trajectory generation, and present two solvers for CPO.

A. Problem Statement

Compared to CIMPC methods, CPO must additionally consider the guidance from the high-level module and generate the corresponding executable trajectory. Based on (6), we define CPO as the single shooting problem:

$$\begin{aligned}
& \min_{\mathbf{u}_{0:N-1}} \sum_{i=0}^{N-1} \ell_{\text{CPO}}(\mathbf{x}_k, \mathbf{u}_k, \hat{\mathbf{p}}_r^*) + V_f(\mathbf{x}_N) \\
& \text{s.t. } \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, T-1, \\
& \mathbf{x}_k \in \mathcal{X}, \quad k = 0, \dots, T, \\
& \mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, T-1.
\end{aligned} \tag{34}$$

Here $\hat{\mathbf{p}}_r^*$ is explicitly included in the objective function. We next investigate two key questions: in what form should contact prior guide contact planning, and how should a feasible approach and contact trajectory be defined mathematically?

1) *Role of Contact Guidance*: In the previous section, we discussed how CIMPC methods suffer from gradient sparsity and proposed CSO, which serves as a pointwise global search on the object and provide contact guidance. But this raises another open question: in what form should the guidance affect contact execution? The most straightforward approach is full coupling, where the contact planning strictly follows and executes the guidance from the CSO. **However, such a fully coupled scheme may compromise control robustness because the upper-level surrogate abstracts away detailed contact interactions. Moreover, the object-centric search performed by CSO does not explicitly account for robot morphology, resulting in an inherent gap between contact selection and**

execution. Moreover, contact-rich manipulation is inherently underdetermined, as numerous contact-wrench and contact-location trajectories can accomplish the same task. Strictly executing the CSO-optimal trajectory would therefore restrict this intrinsic solution diversity in robot–object interaction. To address this limitation, we introduce an objective-based contact planning module named CPO. Through the ranking strategy, the CPO balances the CSO-optimal contact against nearby and other candidate contacts, establishing a loose coupling between high-level contact selection and low-level contact planning. This design benefits from global-search guidance to escape local optima while preserving the flexibility of contact planning for robust execution.

2) *Contact Switching Trajectory*: Existing CIMPC methods mainly induce nearby contacts through attractive costs, without explicitly modeling the detouring motions required for either approaching designated contacts or switching between contacts. To incorporate the upper-level guidance, we define the approaching trajectory in the contact-free space $\mathcal{S}_{\text{free}}$ as:

- All intermediate waypoints lie in the collision-free space.
- The terminal waypoint establishes contact, and the contact force lies within the friction cone.

As shown in Fig. 5, the robot–object interaction states form a low-dimensional manifold embedded in the full system state space. A detouring motion can be viewed as planning a kinodynamically feasible bridging manifold $\mathcal{S}_{\text{approach}}$ within the contact-free space $\mathcal{S}_{\text{free}}$, connecting two states on the contact manifold $\mathcal{S}_{\text{contact}}$. This is challenging for CIMPC methods because, as discussed above, $\mathcal{S}_{\text{free}}$ provides little informative contact-induced gradient for such transitions. To naturally bridge the approaching, contact, and contact-switching phases without relying on a discrete state machine, we formulate CPO under a unified objective that jointly governs contact execution and pre-contact detouring and approach motions.

B. Ranking Strategy of Contact Locations

In addition to the optimal contact location $\hat{\mathbf{p}}_r^*$ from the CSO prediction, we always consider the nearest point $\mathbf{p}_{\text{near}}$ to the current robot configuration as a candidate contact location:

$$\mathbf{p}_{\text{near}} = \arg \min_{\mathbf{p}_i \in \mathcal{I}_{\text{avail}}} \|\mathbf{p}_{\text{ee}} - \mathbf{p}_i^{\text{ee}}\|_2, \quad (35)$$

where $\mathbf{p}_i^{\text{ee}}$ is the end-effector position corresponding to contact at $\mathbf{p}_i$. We reformulate the problem (34) as tracking the reference contact location $\mathbf{p}_{\text{ref}}$, where $\mathbf{p}_{\text{ref}}$ is generated in real time through a ranking strategy between $\mathbf{p}_{\text{near}}$ and $\hat{\mathbf{p}}_r^*$, i.e., $\mathbf{p}_{\text{ref}} \in \{\mathbf{p}_{\text{near}}, \hat{\mathbf{p}}_r^*\}$. Accordingly, the ranking output determines whether to maintain the current contact or detach and switch to $\hat{\mathbf{p}}_r^*$ in the contact phase, and whether to establish a nearby contact or detour toward $\hat{\mathbf{p}}_r^*$ in the pre-contact phase. We next define the evaluation criteria for the ranking strategy.

For each candidate contact location $\mathbf{p}_i$, the optimal cost reduction obtained from (32) is denoted as $\Delta \ell_t^i = \ell(\mathbf{x}_t) - \ell(\hat{\mathbf{x}}_t^+)$. Based on the predicted cost reduction range of all candidate contact locations, $[\Delta \ell_t^{\min}, \Delta \ell_t^{\max}]$, the relative ranking score of the nearby contact $\mathbf{p}_{\text{near}}$ is defined as

$$\rho_t = \Pi_{[0,1]} \left(\frac{\Delta \ell_t^{\text{near}} - \Delta \ell_t^{\min}}{\max(\Delta \ell_t^{\max} - \Delta \ell_t^{\min}, \epsilon)} \right), \quad (36)$$

Algorithm 2 Ranking Strategy of CPO

Input: $\hat{\mathbf{p}}_r^*$ predicted by CSO, current robot configuration $\mathbf{q}_k$, threshold δ

Output: Selected contact location $\mathbf{p}_{\text{ref}}$ and ℓ_{lp}

- 1: Compute nearest point $\mathbf{p}_{\text{near}}$ to end-effector via (35)
 - 2: Evaluate contact cost ℓ_{near} via (30)
 - 3: Compute γ via (36), (37), (38) ▷ Ranking strategy
 - 4: **if** $\gamma = 0$ **then**
 - 5: $\mathbf{p}_{\text{ref}} = \hat{\mathbf{p}}_r^*$, $\ell_{\text{lp}} = \ell_{\text{lift}}$ ▷ Lifting phase
 - 6: **else**
 - 7: $\mathbf{p}_{\text{ref}} \leftarrow \mathbf{p}_{\text{near}}$, $\ell_{\text{lp}} = \ell_{\text{place}}$ ▷ Placing phase
 - 8: **end if**
-

where the fraction performs min-max normalization and $\Pi_{[0,1]}$ projects the resulting score onto $[0, 1]$. The ranking score ρ_t indicates the relative improvement of the nearby contact compared with all candidate contacts.

Next, we define a binary trigger variable $\kappa_t \in \{0, 1\}$ to determine whether the nearby contact is suitable for replacing the CSO-selected contact:

$$\kappa_t = \mathbf{1}(\rho_t \geq \bar{\rho}), \quad (37)$$

where $\bar{\rho}$ is a predefined ranking threshold and $\mathbf{1}(\cdot)$ denotes the indicator function. When $\kappa_t = 1$, the nearby contact $\mathbf{p}_{\text{near}}$ provides a sufficiently competitive cost reduction and can serve as a feasible reference contact. Otherwise, the reference remains the CSO-selected contact $\hat{\mathbf{p}}_r^*$. The trigger variable κ_t therefore evaluates the feasibility of the nearby contact and determines whether contact relocation is allowed.

To avoid frequent switching due to transient ranking variations, we apply a temporal hysteresis strategy to filter κ_t when updating the reference contact.

$$\gamma_{t+1} = \begin{cases} 0, & \kappa_t = 0, \quad t_{(\kappa=0)} \geq T_1, \\ 1, & \kappa_t = 1, \quad t_{(\kappa=1)} \geq T_2, \quad t_{\text{contact}} \geq T_3, \\ \gamma_t, & \text{otherwise,} \end{cases} \quad (38)$$

where $t_{(\kappa=0)}$ and $t_{(\kappa=1)}$ denote the consecutive time counters for the two trigger conditions, and t_{contact} represents the duration of the current contact state. The filtered trigger γ_t is subsequently used in the CPO objective formulation to determine whether the nearby contact or the CSO-selected contact should be prioritized.

The overall ranking process repeatedly evaluates the nearby contact under the current configuration. When $\gamma_t = 1$, the nearby contact $\mathbf{p}_{\text{near}}$ is selected as the reference contact $\mathbf{p}_{\text{ref}}$; otherwise, the robot continues moving toward the guidance $\hat{\mathbf{p}}_r^*$ generated by CSO. The complete procedure is summarized in Algorithm 2. This hierarchical design establishes a weak coupling between CSO and CPO, allowing CPO to exploit the global search guidance from CSO while retaining the flexibility to autonomously determine feasible contact strategies.

C. Formulation of CPO

Contact planning inherently consists of multiple phases, including approaching, contact establishment, and contact

switching. However, the CIMPC formulation does not naturally support contact switching, because contact switching requires leaving $\mathcal{S}_{\text{contact}}$ and traversing the contact-free space $\mathcal{S}_{\text{free}}$, where the CIMPC objective provides no effective gradient information for trajectory optimization. To address this issue, we introduce a novel objective formulation that enables multi phases planning within a unified framework. The proposed formulation is general to different task settings and compatible with various optimization solvers.

As illustrated in Fig. 6, the trajectory optimization is decomposed into three stages. First, the end-effector leaves the current contact and moves around the object toward an intermediate waypoint $\mathbf{p}_{\text{lift}}$ or pose $\mathbf{q}_{\text{lift}}$. This intermediate target guides the robot to reach a collision-free pre-contact configuration. Second, the end-effector is guided toward $\mathbf{p}_{\text{ref}}$ to establish contact. This design is inspired by footstep planning in legged robots [39]. The corresponding cost formulations are introduced below.

1) *Lifting Phase*: During the lifting phase, when the current nearest contact point $\mathbf{p}_{\text{near}}$ is infeasible, a hybrid potential field guides the robot to detach from the object and move above the guidance. It consists of a repulsive term from the object:

$$\ell_{\text{obs}} = \begin{cases} -\log(\|\mathbf{p}_{\text{ee}} - \mathbf{p}_o\|^2 + \varepsilon), & \text{if } \|\mathbf{p}_{\text{ee}} - \mathbf{p}_o\|^2 < \sigma, \\ 0, & \text{otherwise.} \end{cases} \quad (39)$$

and an attractive term toward the lifting point $\mathbf{p}_{\text{lift}}$ or lifting configuration $\mathbf{q}_{\text{lift}}$. The lifting target satisfies: $\|\mathbf{p}_{\text{lift}} - \hat{\mathbf{p}}_r^*\|_2 = d$ or $\|\text{FK}(\mathbf{q}_{\text{lift}}) - \hat{\mathbf{p}}_r^*\|_2 = d$, which guides the robot toward a position above $\hat{\mathbf{p}}_r^*$. The attractive potential is defined as:

$$\ell_{\text{att}} = \|\mathbf{p}_{\text{ee}} - \mathbf{p}_{\text{lift}}\|^2 \quad \text{or} \quad \ell_{\text{att}} = \|\mathbf{q}_{\text{ee}} - \mathbf{q}_{\text{lift}}\|^2. \quad (40)$$

The lifting cost is formulated as:

$$\ell_{\text{lift}} = w_{\text{att}} \ell_{\text{att}} + w_{\text{obs}} \ell_{\text{obs}}. \quad (41)$$

This potential formulation explicitly encodes the lifting behavior and avoids optimization failure caused by gradient-sparse regions during contact breaking.

2) *Placing Phase*: During the placing phase, when the robot reaches a position above $\hat{\mathbf{p}}_r^*$, or when $\mathbf{p}_{\text{near}}$ is verified as a feasible contact location, the end-effector is guided to establish contact with the object. The placing phase is formulated by:

$$\ell_{\text{place}} = w_{\text{con}} \ell_{\text{con}} + w_o \ell_o, \quad (42)$$

where ℓ_{con} attracts the end-effector toward the reference contact point $\mathbf{p}_{\text{ref}}$, and ℓ_o attracts it toward the object center:

$$\ell_{\text{con}} = \|\mathbf{p}_{\text{ee}} - \mathbf{p}_{\text{ref}}\|^2, \quad \ell_o = \|\mathbf{p}_o - \mathbf{p}_{\text{ee}}\|^2. \quad (43)$$

Using only ℓ_{con} may fail to establish physical contact when the object mesh is inaccurate, since perception or projection errors can cause $\hat{\mathbf{p}}_r^*$ to deviate from the actual object surface. The additional term ℓ_o improves robustness against such errors.

3) *Switching Parameter*: The switching between lifting and placing behaviors is determined by the trigger γ introduced in Section IV-B. The overall lift-and-place cost is:

$$\ell_{\text{lp}} = (1 - \gamma)\ell_{\text{lift}} + \gamma\ell_{\text{place}}, \quad (44)$$

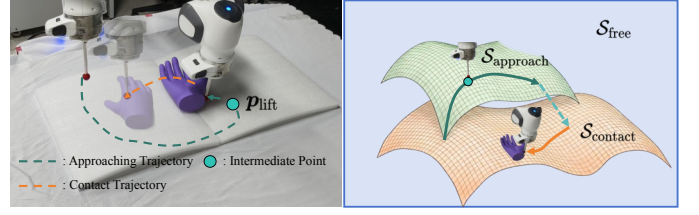


Fig. 5. (a) Illustrative example of a feasible approach and contact trajectory. (b) Explanation of CPO in a geometric perspective. CPO can be characterized as operating over two manifolds: contact states in $\mathcal{S}_{\text{contact}}$ and approaching trajectories on the manifold $\mathcal{S}_{\text{approach}}$ embedded in the contact-free state space $\mathcal{S}_{\text{free}}$. CPO must not only generate contact trajectories on $\mathcal{S}_{\text{contact}}$ but also address the gradient-sparse issue and generate trajectories on $\mathcal{S}_{\text{approach}}$.

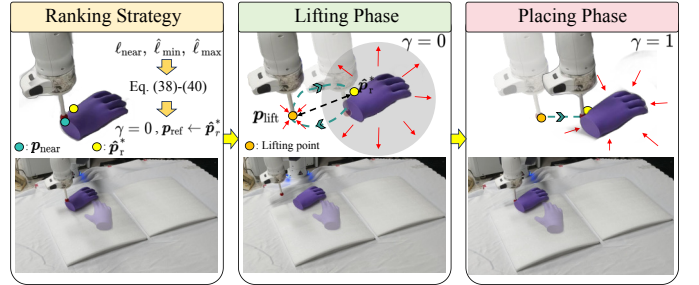


Fig. 6. Illustration of the CPO-based contact planning module. (a) Ranking strategy. When $\mathbf{p}_{\text{near}}$ is evaluated infeasible, γ turns to zero and $\mathbf{p}_{\text{ref}} \leftarrow \hat{\mathbf{p}}_r^*$. The evaluation runs continuously in real-time. (b) When $\gamma = 0$, the robot is guided by the potential field to $\mathbf{p}_{\text{lift}}$. (c) Once above $\hat{\mathbf{p}}_r^*$ or the nearest point $\mathbf{p}_{\text{ref}}$ to the robot is detected feasible via (30), γ becomes 1 and an attractive potential is applied, guiding the robot to contact with the object.

where ℓ_{lp} is incorporated into the stage cost ℓ_{cpo} in (34). By defining the terminal cost V_f identical to (7), the CPO formulation is obtained as:

$$\begin{aligned} \min_{\mathbf{u}_k} \quad & \sum_{i=0}^{N-1} \left(w_{\text{lp}} \ell_{\text{lp}}(\mathbf{x}_k, \mathbf{q}_k, \mathbf{u}_k, \mathbf{p}_{\text{ref}}) + w_u \|\mathbf{u}_k\|^2 \right) + w_{\text{cso}} \ell_{\text{cso}} \\ \text{s.t.} \quad & \mathbf{x}_{k+1} = f_{\text{cf}}(\mathbf{x}_k, \mathbf{u}_k), \quad k = 0, \dots, T-1, \\ & \mathbf{x}_k \in \mathcal{X}, \quad k = 0, \dots, T, \\ & \mathbf{u}_k \in \mathcal{U}, \quad k = 0, \dots, T-1. \end{aligned} \quad (45)$$

The formulation in (45) preserves the favorable properties of the original CIMPC objective while enabling hybrid contact planning. Therefore, CPO can be interpreted as a prior-guided contact planning framework. We state the following lemma.

Lemma 4 (Compatibility of ℓ_{lp} and ℓ_{cso}). ℓ_{lp} and ℓ_{cso} are non-conflicting objectives in CPO: minimizing the lift-and-place objective does not obstruct first-order improvement of the object manipulation objective.

Proof. See Appendix C.

D. Solvers of CPO

Since CPO specifies the optimization formulation rather than a specific solver, various reactive motion planners can be adopted to generate contact trajectories. Below, we present two solvers employed in our experiments.

1) *Operational-Space Control*: We simplify the full-order contact planning problem to an end-effector point contact planning problem, avoiding high-DoF robot trajectory optimization. The resulting joint torques are generated by operational space control (OSC) [40] to track the CPO output. Specifically, the state in (45) is reduced to $\mathbf{s} \in \mathbb{R}^{10}$, consisting of the object position, object orientation represented by a quaternion, and the end-effector position $\mathbf{p}_{ee} \in \mathbb{R}^3$. Solving (45) yields the control input $\mathbf{u} = \Delta \mathbf{p}_{ee}$.

Given the desired pose $(\mathbf{p}_d, \mathbf{R}_d)$ with $\mathbf{p}_d = \mathbf{p}_{ee} + \mathbf{u}$ and a preset $\mathbf{R}_d$, the position and orientation errors are computed as $\mathbf{e}_{\text{pos}} = \mathbf{u}$, and $\mathbf{e}_{\text{R}} = \mathbf{R}_{\text{ee}}^\top \mathbf{R}_d$. Converting $\mathbf{e}_{\text{R}}$ to quaternion form gives $\mathbf{e}_{\text{quat}} = [\epsilon_e, \eta_e]^\top \in \mathbb{R}^4$, and after sign correction,

$$\mathbf{e}_{\text{quat}} \leftarrow \begin{cases} \mathbf{e}_{\text{quat}}, & \eta_e \geq 0, \\ -\mathbf{e}_{\text{quat}}, & \eta_e < 0, \end{cases} \quad \mathbf{e}_{\text{rot}} = -\mathbf{R}\epsilon_e. \quad (46)$$

By defining $\mathbf{e}_{\text{pose}} = [\mathbf{e}_{\text{pos}}, \mathbf{e}_{\text{rot}}]^\top$, the task-space and null-space torques are given by:

$$\boldsymbol{\tau}_{\text{task}} = \mathbf{J}^\top(\mathbf{q}) (-\mathbf{K}_x \mathbf{e}_{\text{pose}} - \mathbf{D}_x \dot{\mathbf{x}}), \quad (47)$$

$$\boldsymbol{\tau}_{\text{null}} = (\mathbf{I} - \mathbf{J}^\top(\mathbf{J}^\top)^\dagger) (\mathbf{K}_q(\mathbf{q}_{\text{init}} - \mathbf{q}) - 2\sqrt{\mathbf{K}_q} \dot{\mathbf{q}}), \quad (48)$$

and the final control torque is $\boldsymbol{\tau} = \boldsymbol{\tau}_{\text{task}} + \boldsymbol{\tau}_{\text{null}} + \mathbf{g}(\mathbf{q})$. This controller is used in all simulations and real-world experiments in Section VI, and is particularly suitable for tasks such as tabletop manipulation, where high orientation diversity is not critical.

2) *Sampling-based Motion Planner*: Another formulation is to directly solve the full-order planning problem of the robot. In this case, the state in (45) is given by $\mathbf{x} \in \mathbb{R}^{7+n_{\text{DoF}}}$, where n_{DoF} denotes the number of robot DoFs. For this class of problems, we employ sampling-based MPC methods as the solver, such as MPPI [41]. Recent studies have demonstrated the impressive performance of MPPI in solving high-dimensional optimization problems for redundant robots, enabling real-time computation while avoiding issues related to gradient-based methods. The cost function is defined in the same manner as in the previous formulation, except that $\mathbf{p}_{ee}$ is computed via forward kinematics for the full-order problem.

V. EVALUATION OF SURROGATE CONTACT MODEL

This section validates the fidelity of SCM before system-level experiments. SCM replaces the dense Delassus matrix $\mathbf{W}_{\text{env}}$ by its block-diagonal approximation and evaluates $\hat{\boldsymbol{\lambda}}_{\text{env}}$ in closed form under Assumptions 1–2. We verify that the resulting one-step prediction error against higher-fidelity models is small, and that occasional large single-step gaps do not prevent task success when the CSO-optimal force is applied directly.

We use the fingertip ground-rotation setup with the same $h = 0.02$ s, $\mathbf{M}_o$, and success condition (49) as below. At each step, CSO selects the optimal candidate $\hat{\mathbf{p}}_{\text{r}}^* \in \mathcal{I}_{\text{avail}}$ and its best force $\hat{\boldsymbol{\lambda}}_{\text{r}}^*$, and SCM predicts $\hat{\mathbf{v}}^+$ and $\hat{\boldsymbol{\lambda}}_{\text{env}}$. Under identical free velocity $\mathbf{v}_{\text{free}}$, environment Jacobian $\tilde{\mathbf{J}}_{\text{env}}$, and $\mathbf{M}_o^{-1}$, we compute the rigid-pyramid LCP reference via Lemke’s algorithm and a MuJoCo QP reference via a single `mj_forward`.

TABLE II
FIDELITY OF SCM AGAINST THE RIGID LCP AND SUCCESS OF DIRECT-APPLY ROLLOUTS.

Metric	Success	dir θ (rad)	mag. gap	$\ d\boldsymbol{\lambda}\ _D$
SCM vs. LCP (median)	—	0.19	0.005	0.05
SCM vs. LCP (mean)	—	0.33	0.029	0.069
Direct-apply rollout	100%	—	—	—

Following Lemma 3 and Appendix B, we report the inertia-weighted direction error θ with $\cos \theta_{M_o}$, the magnitude gap $|\|\mathbf{v}^+\|_{M_o} - \|\hat{\mathbf{v}}^+\|_{M_o}|$, and the multiplier gap $\|\hat{\boldsymbol{\lambda}}_{\text{env}} - \boldsymbol{\lambda}_{\text{env}}\|_D$. Each rollout then directly applies the CSO-optimal contact by tracking the predicted orientation and the goal-directed component of the predicted translation, without CPO or OSC.

Table II summarizes the results over repeated ContactDB objects. Against LCP, the median direction error is around 0.19 rad with magnitude and multiplier gaps at the 10^{-3} – 10^{-2} scale; larger gaps concentrate on a few contact-set transition steps. The MuJoCo single-step gap is occasionally larger due to its full-body contact resolution, but the goal-directed motion trend is preserved. Every direct-apply trajectory satisfies (49) within 160 steps, indicating that SCM preserves the contact-selection signal needed by CSO.

VI. EXPERIMENT: NONPREHENSILE MANIPULATION

This section evaluates the SCSP framework through experiments. We first conduct nonprehensile manipulation experiments, which represent a complex contact-rich manipulation task with high demands on operational diversity and control robustness. The task is to adjust the pose of an object on the ground through nonprehensile manipulation, which consists of:

- **On-ground rotation**: The robot needs to move the object to the target pose on the ground through pushing and rotating, which constitutes a 2D planar pushing task.
- **On-ground flipping**: The robot needs to move the object to the target pose on the ground through pushing and flipping, which constitutes a 6D manipulation task.

We select a set of complex-shaped objects from the ContactDB dataset [42] for validation. The objects are initialized at random poses with $x_{\text{init}}, y_{\text{init}} \in [-0.3, 0.3]$ m, and Euler angles $(\phi_{\text{init}}, \theta_{\text{init}}, \psi_{\text{init}})$ uniformly sampled from $[-\pi, \pi]$ rad. We define the conditions of task success as:

$$\|\mathbf{p}_o - \mathbf{p}_{\text{goal}}\| \leq 0.02 \text{ [m]}, \quad 1 - (\boldsymbol{\psi}_{\text{goal}}^\top \boldsymbol{\psi}_o)^2 \leq 0.015, \quad (49)$$

where the orientation tolerance corresponding to the quaternion error threshold is approximately 14.1° . A trial is considered successful if the conditions are satisfied within 2500 rollout steps. We evaluate the algorithm on different robot platforms, namely a fingertip³ and the Franka Emika Panda, and compare different methods in terms of success rate, efficiency, and convergence error. **These experiments evaluate whether SCSP can select contact locations and complete tasks under complex contact interactions.**

³The “fingertip” is a floating point with a collision volume and is commonly used as a simplified setup for validating contact planning methods.

Simulations are conducted in `MuJoCo` [43] and `IsaacGym` [44] with a fixed integration time step $h = 0.02s$. For the sampling method in our CSO module, we use `CoACD` [45] for convex decomposition of the objects and `trimesh` for mesh processing and local frame construction. The inner loop of CSO is implemented by solving the nonlinear SQP optimization using `CasADi` [46] with `SNOPT` [47] solver, and the solver of CPO is `IPOPT` [48]. All experiments are conducted on a desktop computer with the Intel i7-13620H processor and 32 GB RAM.

We applied certain randomization to the environment and object parameters. The friction coefficients of the object μ_o , fingertip μ_r , and ground μ_{env} are all set to 0.5 with random perturbations of ± 0.2 . The object mass is set to 0.1 kg and randomly sampled from the discrete set $\{0.01, 0.1, 1.0, 10.0\}$ kg. We set $M_o = \text{diag}(50, 50, 50, 0.05, 0.05, 0.05)$ and $K_r = 300I_7$, and the dynamic parameters used in SCM are not measured precisely, but are instead roughly set to their mean values. For CSO, we set $n_s = 70$ and $\lambda_{\max}^n = 0.2$. For CPO, we set $\bar{\rho} = 0.75$, $T_1 = 5$, $T_2 = 10$, $T_3 = 15$, $\varepsilon = 0.01$, $w_{lp} = 1$, $w_{pos} = 500$, $w_{quat} = 5$, $w_u = 50$, $w_{obj} = 10$. We conducted parameter tuning via grid search over 100 randomly sampled trials, and the resulting parameters were fixed for all subsequent experiments. In addition, based on the definition of the object's local coordinate frame and the planar environment in the experiments, the stable pose set $\mathcal{X}_g$ can be written as $\mathcal{X}_g = \{\mathbf{x}_o \in \mathcal{X} \mid \mathbf{R}(\mathbf{x}_o)\mathbf{e}_z = \mathbf{e}_z\}$ with $\mathbf{e}_z = [0, 0, 1]^\top$. Therefore, ℓ_{stab} is defined as:

$$\ell_{\text{stab}} = 1 - (\mathbf{R}(\mathbf{x}_o)\mathbf{e}_z)^\top (\mathbf{R}(\mathbf{x}_{\text{goal}})\mathbf{e}_z). \quad (50)$$

From Table I, we select the methods that are applicable to 6D manipulation, operate in real time, and can handle arbitrary manipulable object geometries as baselines, including:

- **DyWA** [17]: An end-to-end 6D nonprehensile manipulation with generalization across diverse unseen objects. We follow the settings in their released source code by adding the test objects to the `IsaacGym`, and evaluate the performance of the single-view student policy.
- **Sampling** [25]: A class of hierarchical frameworks in which the upper layer selects a set of candidate contact points by sampling on the object envelope. The upper layer performs contact selection by evaluating the CIMPC cost and control cost in parallel, while the lower layer uses OSC to achieve approach and contact. To eliminate the effect of differences in contact modeling, we replace the original C3 model [18] in [25] with f_{cf} .
- **I-MPC**: A CIMPC method based on a QP-based contact model. This method performs contact simulation by solving the KKT conditions of the contact model and implicitly plans the contact mode, force, and location by optimizing robot-centric control inputs. The stage cost is defined as the sum of a contact cost and a control cost $\ell = \|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{ec}}\|^2 + \|\mathbf{u}\|^2$ and the terminal cost is defined the same as (7). Although the contact planning formulation is unified and does not use hierarchical solving for different contact modes, the planning horizon is short to enable online computation. In addition, the method relies on explicitly defined contact costs to encourage

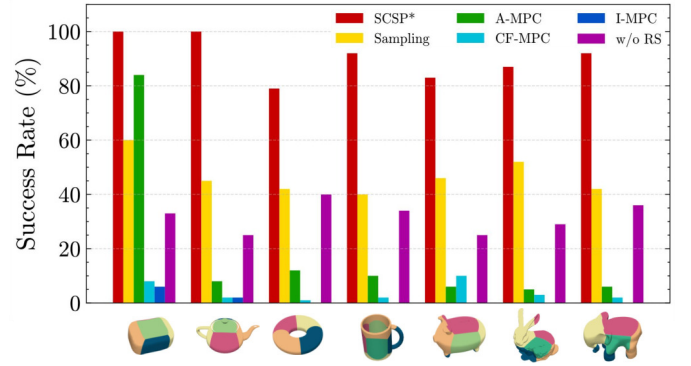


Fig. 7. Experimental results of fingertip manipulation. SCSP achieves a substantially higher success rate than the other methods.

robot-object interaction. As a result, it can only adjust the contact location to a limited extent.

- **CF-MPC** [22]: A complementarity-free MPC method based on closed-form contact model (12). The cost design is the same as in I-MPC, but it uses an improved contact model, achieving higher manipulation success rates and better computational efficiency. However, its underlying formulation remains unchanged, so its contact selection capability is still limited.
- **A-MPC**: We augment CF-MPC by adding an additional alignment cost into the stage cost:

$$\ell_{\text{align}} = -\frac{(\bar{\mathbf{v}}_{\text{obj}}^{\text{goal}})^\top \bar{\mathbf{v}}_{\text{ec}}^{\text{obj}} + 1}{2}, \quad (51)$$

in which: $\bar{\mathbf{v}}_{\text{obj}}^{\text{goal}} = \frac{\mathbf{p}_{\text{goal}} - \mathbf{p}_{\text{obj}}}{\|\mathbf{p}_{\text{goal}} - \mathbf{p}_{\text{obj}}\|}$, $\bar{\mathbf{v}}_{\text{ec}}^{\text{obj}} = \frac{\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{ec}}}{\|\mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{ec}}\|}$. This cost is motivated by a simple intuition: when pushing an object toward a given direction, the optimal contact location often lies on the side opposite to the pushing direction, since the surface is more likely to contain the desired force within the friction cones. Although this heuristic may not hold for objects with complex geometries, it serves as a simple prior that can guide contact planning methods to escape local minima.

- **w/o RS**: A variant of SCSP without the Ranking Strategy in CPO. In this method, CSO and CPO are fully connected, which means that CPO completely follows and executes the contact locations predicted by CSO. This ablation demonstrates that the ranking strategy is a crucial factor for the robustness of SCSP.

A. Fingertip Manipulation in Simulation

We first conduct fingertip manipulation to validate the critical role of contact selection in contact-rich manipulation. Since DyWA is an offline-trained policy, it is not applicable to the fingertip experiments and is therefore omitted. By comparing with other model-based methods, we demonstrate that explicitly introducing a contact selection module is effective for handling complex manipulation tasks.

The experimental results are shown in Fig. 7, 9 and Table III. Compared with the baseline methods, SCSP achieves a significantly higher success rate, shorter execution time, and

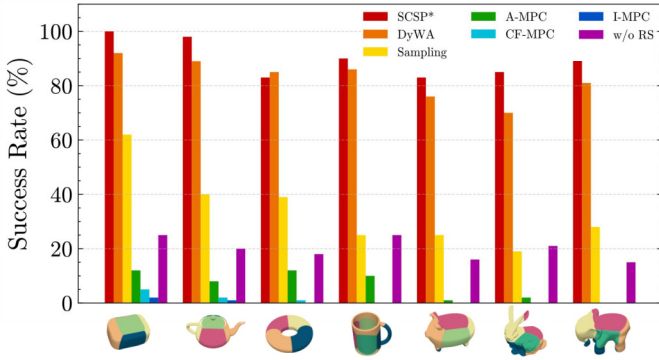


Fig. 8. Experimental results of robot manipulation in simulation. SCSP significantly outperforms other model-based methods and exceeds the performance of well-trained learning-based methods.

lower final position and quaternion errors. Our observations indicate that most of the successful cases of I-MPC and CF-MPC correspond to a good robot initial state, where the methods can succeed by applying contact forces at a fixed contact location. When the fingertip initial position is poor, they exhibit almost no adjustment capability. A-MPC achieves a significant improvement in success rate, demonstrating that even a coarse contact selection prior can substantially mitigate the local optimum problem in contact planning. Sampling achieves further improvement but at the cost of lower computational efficiency. These experimental results highlight the importance of active contact selection in complex tasks.

B. Robot Manipulation in Simulation

We further evaluate SCSP in redundant robot manipulation tasks. Compared with floating-based fingertip, contact problems for redundant robots introduce complex kinematic constraints. In the experiments, we demonstrate the superiority of SCSP and validate the soundness of our framework.

Due to the strong dependencies of the source code released for DyWA, which are difficult to transfer, we follow its setup and conduct all experiments in *IsaacGym* for consistency, with other settings similar to Section VI-A. All methods use our link-mounted sphere as the end-effector to replicate point contacts. Since DyWA uses the Franka gripper, we adjust the link length so that the sphere's center coincides with the gripper's center. The Cartesian stiffness $\mathbf{K}_x = \text{diag}(1000, 1000, 1000, 50, 50, 50)$ and $\mathbf{D}_x = \sqrt{2\mathbf{K}_x}$. All methods use the same settings as in the fingertip experiments, except that $\mathbf{u} \in \mathbb{R}^7$ in I-MPC, CF-MPC and A-MPC. $\mathbf{p}_{ec}$ is obtained via differentiable forward kinematics.

The experimental results are shown in Fig. 8, Fig. 9, and Table III. SCSP achieves a higher success rate than other model-based algorithms and outperforms DyWA on all objects except the torus. Additionally, SCSP achieved the shortest execution time, demonstrating that SCSP can generate correct contact locations through CSO, thereby efficiently minimizing the pose cost of the object. SCSP also achieved the smallest convergence position error and quaternion error. Furthermore, we observed that during experiments, DyWA tends to flip objects by pressing the edges of the top surface,

whereas our method makes more comprehensive decisions regarding whether to lift the object by pushing the bottom or by pressing the top. Although many flipping tasks can be accomplished through pressing, for certain objects (such as elongated objects), the pressing approach requires applying substantial torque to the object and demands sufficiently high coefficients of friction. Based on our observations, the global contact selection mechanism in CSO enables SCSP to exhibit greater diversity in its manipulation behaviors compared with other methods, rather than converging to a single strategy.

In addition, we conducted two ablation studies. CF-MPC and A-MPC can be regarded as ablations of the contact selection strategy. CF-MPC performs contact planning without contact selection, whereas A-MPC replaces the physical reasoning of CSO-based contact planning with a simple prior. Compared with SCSP, both methods show a substantial decrease in success rate, indicating that reasonable contact selection is crucial for complex 6D nonprehensile manipulation tasks. The w/o RS variant serves as an ablation of the ranking strategy. We observed that, since w/o RS fully follows the high-frequency output of CSO, it frequently interrupts ongoing manipulation and switches contact points, resulting in a significant drop in success rate and an increase in execution time. This suggests that it is necessary to comprehensively evaluate the contact locations produced by CSO and to balance smooth execution against the global optimality of the selected contact location.

We further validated the impact of different numbers of sampled candidate points on performance. As shown in Table IV, when the number of sampled points reaches 70, nearly the highest success rate is achieved with relatively low SCSP computation time. As the number of sampled points continues to increase, the success rate plateaus and computational efficiency deteriorates. This indicates that increasing the number of sampled points does not necessarily improve performance. There exists an optimal convergence value. For nonprehensile manipulation tasks, 70 to 100 sampled points are sufficient to characterize the geometric features of object geometries.

C. Ablation Studies

D. Real-world Experiments

Finally, we validate the robustness of SCSP to in real-world experiments. Most model-based methods perform well in simulation, but in real-world experiments, they often rely on highly accurate perception systems, such as motion capture. The main challenge in replacing exact perception with purely vision-based perception is that errors in pose observation can perturb contact estimation, leading to incorrect inference of contact states and consequently inaccurate contact dynamics. To the best of our knowledge, our method is the first model-based approach that can achieve purely vision-based, complex geometries, model mismatch contact-rich manipulation.

1) *Experimental Setup*: We conduct experiments using a Franka Emika Panda equipped with a 3D-printed end-effector, and arbitrarily select 4 irregularly shaped objects: Duck, Bunny, Hand, and Xbox Controller, as shown in Fig. 10. We evaluate on-ground rotation on Duck and Bunny, and on-ground flipping on Bunny, Hand, and Xbox Controller.

TABLE III
COMPARISON OF NONPREHENSILE MANIPULATION TASKS OF THE FINGERTIP EXPERIMENT AND THE ROBOT EXPERIMENT (200 TRIALS).

Platform	Method	Success Rate		Execution Time (s)		Planning Time (s)		Obj. Pos. Error (m)		Obj. Quat. Error	
		Rot.	Flip.	Rot.	Flip.	Rot.	Flip.	Rot.	Flip.	Rot.	Flip.
Fingertip	SCSP*	0.97	0.85	10.09	15.68	0.023	0.023	0.01	0.008	0.016	0.019
	DyWA	/	/	/	/	/	/	/	/	/	/
	Sampling	0.89	0.61	18.53	18.56	0.205	0.246	0.015	0.016	0.018	0.020
	I-MPC	0.05	0.03	19.62	24.89	0.032	0.034	0.016	0.017	0.037	0.037
	CF-MPC	0.07	0.05	18.15	24.83	0.006	0.006	0.018	0.012	0.025	0.027
	A-MPC	0.50	0.18	12.21	19.72	0.012	0.068	0.015	0.016	0.022	0.021
	w/o RS	0.55	0.14	19.42	23.78	0.024	0.026	0.019	0.019	0.037	0.042
Franka	SCSP*	0.95	0.81	15.16	18.21	0.024	0.025	0.009	0.012	0.015	0.021
	DyWA	0.86	0.69	18.12	20.56	0.031	0.031	0.015	0.016	0.019	0.021
	Sampling	0.56	0.39	18.97	19.68	0.228	0.252	0.014	0.012	0.016	0.022
	I-MPC	0.01	0.00	22.62	/	0.083	0.085	0.018	/	0.036	/
	CF-MPC	0.01	0.01	21.18	22.56	0.082	0.081	0.016	0.018	0.026	0.027
	A-MPC	0.10	0.01	18.32	23.72	0.019	0.028	0.016	0.016	0.021	0.020
	w/o RS	0.36	0.18	19.36	21.57	0.023	0.025	0.019	0.018	0.031	0.036

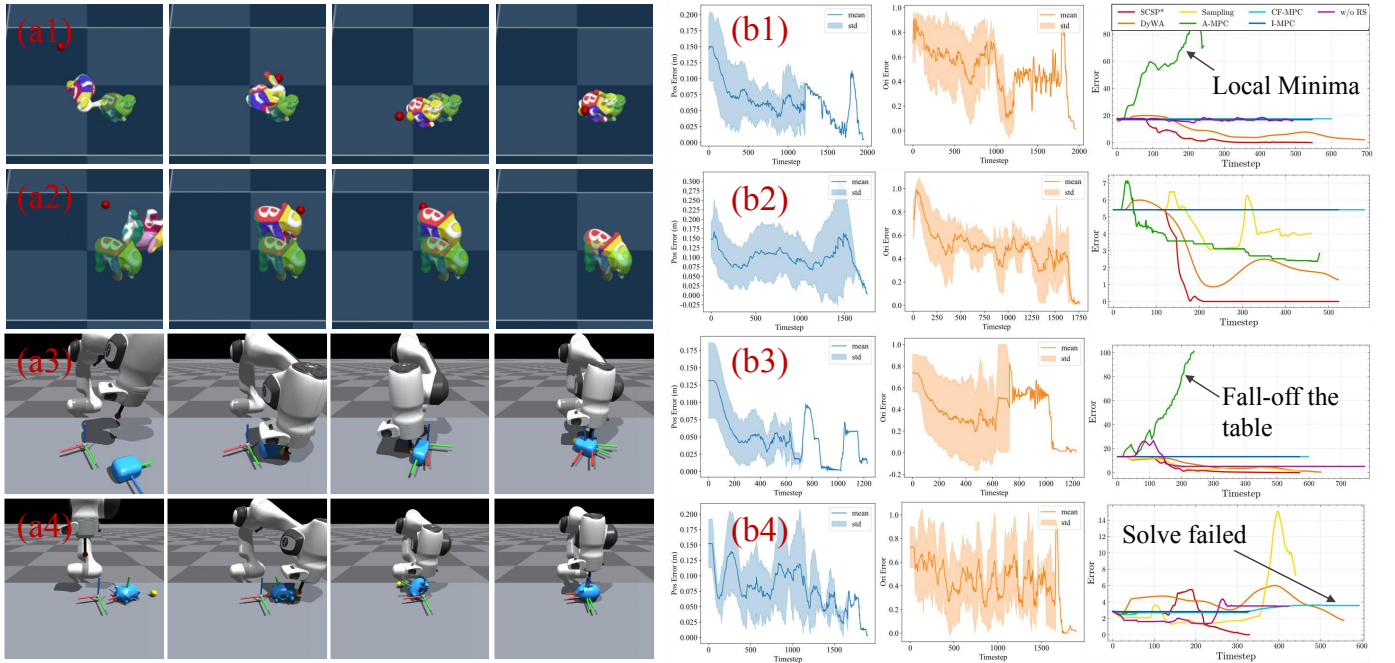


Fig. 9. Nonprehensile manipulation experiments in simulation. (a1)-(a4): Snapshots of SCSP in the fingertip and robot experiments, where the goal pose is visualized either transparently or as a floating coordinate frame. (b1)-(b4): The left two panels show the mean and standard deviation curves of the position and orientation errors over steps across 10 SCSP trials. The right panel shows the pose error ℓ_{pose} of different methods during manipulation. The comparison methods either require longer execution time or fail due to local minima or solver failures.

TABLE IV
EVALUATION OF THE EFFECT OF THE NUMBER OF CANDIDATE POINTS

Sampling numbers in CSO	5	10	70	120	500
Avg. Success Rate	0.25	0.59	0.88	0.88	0.85
SCSP Computation Time (s)	0.012	0.017	0.025	0.032	0.089

The object CAD models are obtained from the ContactDB dataset and online repositories. To validate the robustness of our method under inaccurate dynamics, we provide only

rough estimates of the object's physical parameters such as mass and friction coefficients. Moreover, instead of relying on precise motion capture for exact perception, we perform state estimation by FoundationPose [49] using a single-view camera (Realsense D435i). In real-world experiments, precise object-environment contact locations are not directly observable. To address this, we approximately reconstruct the task environment in MuJoCo for contact estimation, as shown in Fig. 11. We perform a simple calibration procedure to determine the object's scale parameter and positional offsets in the simulator to eliminate contact estimation misclassification.

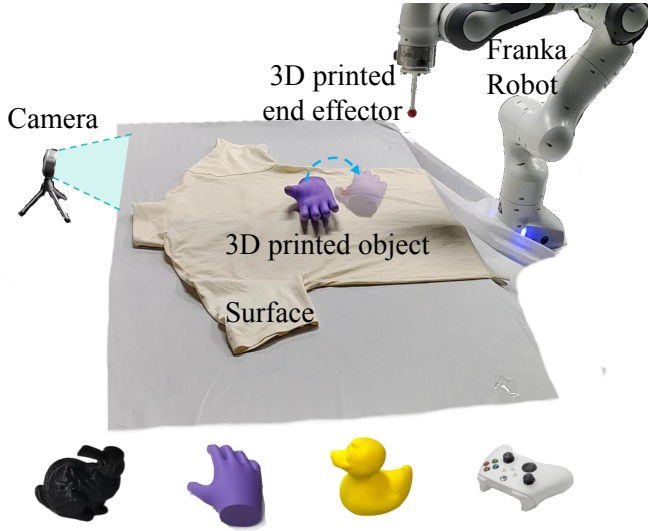


Fig. 10. Hardware setup of the real-world experiments.

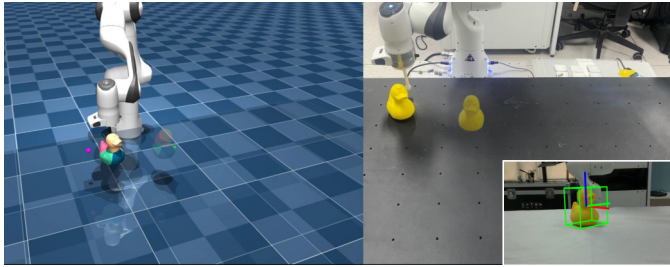


Fig. 11. Perception method in the real-world experimental setting. Object pose is detected using *FoundationPose* and filtered via a Kalman filter. Contact estimation is performed using a *MuJoCo* implementation that approximates the experimental environment.

Specifically, we use Franka’s kinesthetic teaching mode to move the end-effector and make contact with several randomly selected locations on the object, and then tune the object’s scale and the offsets $[\Delta x, \Delta y, \Delta z]$ such that the simulator can approximately predict whether the end-effector is in contact with the object. The goal pose range, success conditions, and maximum execution steps for the task are identical to those in the simulation experiments. Sampling and DyWA use the same settings as in the previous section, and the outputs of DyWA are executed using an impedance controller.

The OSC controller is implemented in *C++*. Unlike in simulation experiments, control in real-world deployment is subject to disturbances, leading to control lag and steady-state errors. To address this, we designed an adaptive controller for precise control, which we refer to as the Incremental OSC. We introduce an intermediate position vector $\mathbf{p}_{\text{inc}} = \mathbf{p}_{\text{inc}} + \mathbf{K}_{\text{inc}}\mathbf{u}$, and replace $\mathbf{e}_{\text{pos}}$ with $\mathbf{e}_{\text{pos}} = \mathbf{p}_{\text{inc}} - \mathbf{p}_{\text{ee}}$. The steady-state error is adaptively eliminated under the high-frequency self-integrating effect. In the experiment, we set $\mathbf{K}_{\text{inc}} = 0.01\mathbf{I}_3$, $\mathbf{K}_x = \text{diag}(500, 500, 500, 10, 10, 10)$ and $\mathbf{D}_x = \sqrt{2\mathbf{K}_x}$.

2) *Results and Discussion*: We first evaluate the success rate of SCSP across different objects and tabletop materials to assess its robustness under model inaccuracies. The experimental results are shown in Fig. 12, 13 and Table V. They

indicate that our method maintains strong performance even under perception noise and model mismatch.

We conducted each experiment on two different tabletop materials: smooth and rough as in Fig. 14. For the on-ground rotation experiments, the smooth surface denotes a tabletop with low friction, whereas the rough surface denotes cotton clothes. For the on-ground flipping experiments, which require greater friction, we use foam padding as the rough surface and cotton cloth as the smooth surface. Note that our method does not measure friction coefficients, object mass, or other physical parameters, but instead uses coarse estimates. Additionally, biases in contact estimation and noise in object pose estimation can interfere with the decision-making of SCSP. The experiment results suggest that SCSP can effectively cope with model inaccuracies and disturbances, which is typically challenging for model-based methods.

As shown in Table V, SCSP achieved a 100% success rate in the on-ground rotation experiments with complex object geometries and also showed strong performance in the flipping tasks, whereas the comparison methods failed in nearly all flipping experiments. We observe that the failures of DyWA mainly arose from contact slippage and violations of the Franka acceleration limits during pressing, while the failures of Sampling were primarily caused by the selection of unstable contact locations. Due to the complex geometry and relatively smooth material of the 3D printed objects, the fingertip sometimes fails to establish stable contact with the object, resulting in relative sliding that causes the execution of the contact plan planned by SCSP to fail. Additionally, surface friction also affects the success rate of flipping. As shown in the results, SCSP achieves a notably higher success rate on the rough surface compared with the smooth surface.

We further conduct continuous manipulation experiments, as shown in Fig. 13. During manipulation, we repeatedly perturb the object pose to evaluate the real-time performance, closed-loop planning capability, and robustness of SCSP under accumulated perception errors. Under repeated perturbations, both the pose estimated from *FoundationPose* and the contact estimation become biased. However, SCSP exhibits strong replanning capability and consistently completes the continuous manipulation tasks under disturbances.

We combine the preceding experimental results to analyze the source of SCSP’s robustness to model mismatch. We conclude that: 1) the ablation study reveals that optimality of the contact prior is not necessary. Although the prior is important for contact planning, even a coarse contact location selection as a prior can still bring substantial performance improvement. This observation also provides insight into the role of the contact-selection model: how to properly connect the prior with the planning framework seems to be more important than the optimality and accuracy of the prior itself. The designs of CSO and the ranking strategy are both developed with this perspective. 2) CPO has a strong closed-loop nature. The real-time performance and robustness of CIMPC methods have been thoroughly validated in previous work. CPO is a prior-guided contact planning framework built upon the CIMPC formulation. It inherits the robustness of CIMPC while improving manipulation diversity. Therefore, the SCSP

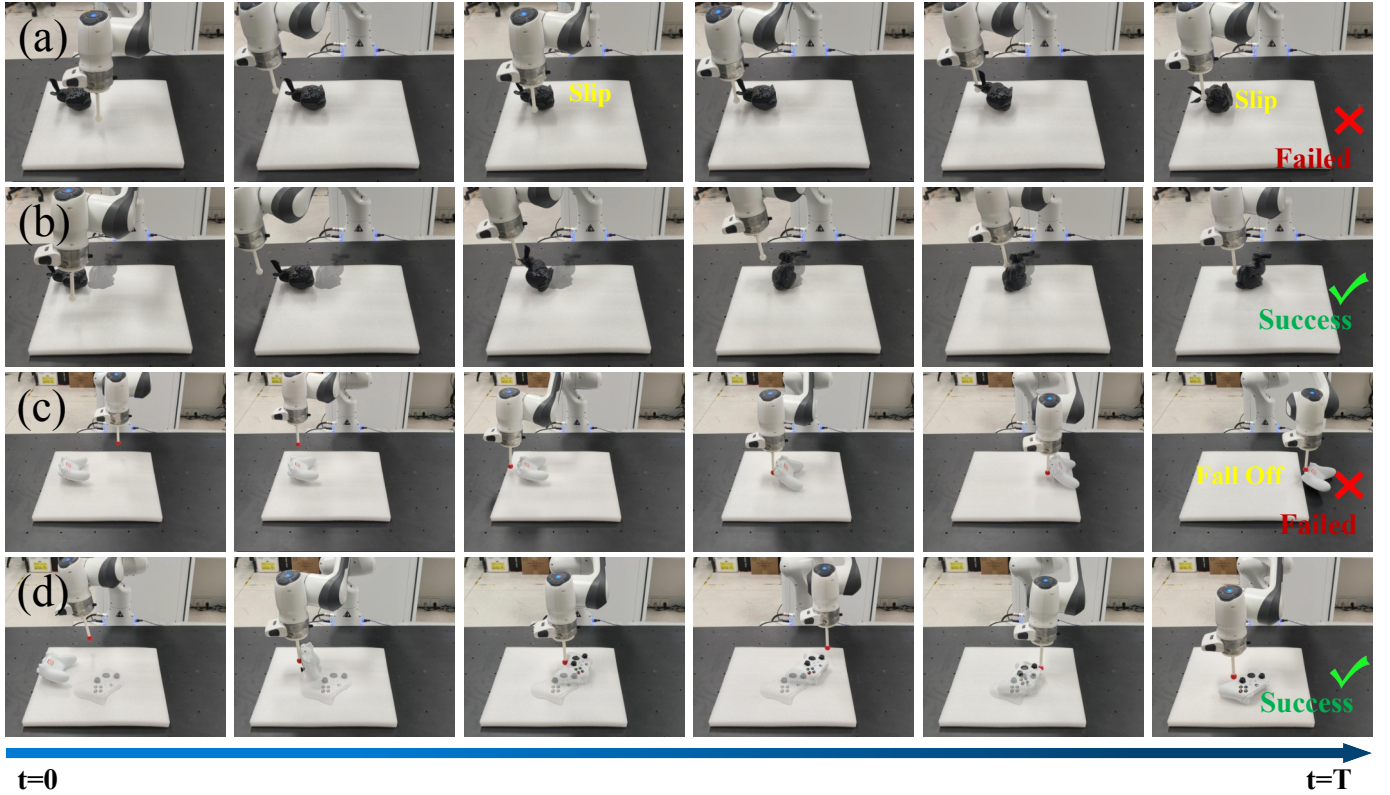


Fig. 12. Snapshots of nonprehensile manipulation in real-world experiments. (a) **DyWA** failed to flip the bunny. (b) **SCSP** succeeds in flipping the bunny. (c) **Sampling** failed by pushing the Xbox controller to fall off the foam padding. (d) **SCSP** succeed in flipping the Xbox controller.

TABLE V
SUCCESS RATES (%) FOR ON-GROUND ROTATION AND FLIPPING ON DIFFERENT SURFACES AND OBJECTS.

Method	On-ground rotation				On-ground flipping					
	Duck		Bunny		Bunny		Hand		Xbox	
	Smooth	Rough	Smooth	Rough	Smooth	Rough	Smooth	Rough	Smooth	Rough
Sampling	9 / 20	5 / 20	7 / 20	4 / 20	0 / 20	0 / 20	0 / 20	0 / 20	0 / 20	0 / 20
DyWA	6 / 20	4 / 20	2 / 20	1 / 20	0 / 20	0 / 20	0 / 20	0 / 20	0 / 20	0 / 20
SCSP*	20 / 20	19 / 20	5 / 20	5 / 20	2 / 20	3 / 20	3 / 20	4 / 20	4 / 20	5 / 20

TABLE VI
COMPUTATION TIME OF SCSP

Metric	Value
CSO Computation time	
Offline sampling cost time (s)	4.82 ± 3.85
Exchange cost time (ms)	0.002 ± 0.01
MIQP cost time (ms)	0.23 ± 0.01
CSO solving time (ms)	16.21 ± 0.58
CPO Computation time	
RS cost time (s)	0.01 ± 0.00
Average CPO solving time (ms)	8.68 ± 0.31

framework is able to achieve robust performance.

VII. GENERALIZATION TO MULTIPLE TASKS

We further implement SCSP in different scenarios to evaluate its generalization. Using multi-contact tasks and long-horizon tasks as examples, we demonstrate that SCSP can

perform **online contact selection** and execute more complex manipulation tasks on more underactuated objects.

A. Case Study I: Multi-Contact Task

SCSP can be easily extended to multi-contact tasks. We illustrate the generalization of SCSP with following example:

1) *Soccer Dribbling*: We use a soccer dribbling task to demonstrate **the performance of SCSP in dynamic scenarios**. In this task, SCSP plans contact with the ball using one of the two feet and drives it toward the goal through multi-contact manipulation. The main challenges lie in the highly dynamic behavior of the ball, which makes stable and sustained contact difficult, and in coordinating the two feet effectively. We read the ball position p_k and estimate its velocity v_k from the simulation using a Kalman filter. The 3D model of the ball is then input to the SCSP to compute the actions $u \in \mathbb{R}^{2 \times 4}$ for both feet, where the control vector for each foot is $[x, y, z, \theta_{yaw}]$. We define activation triggers

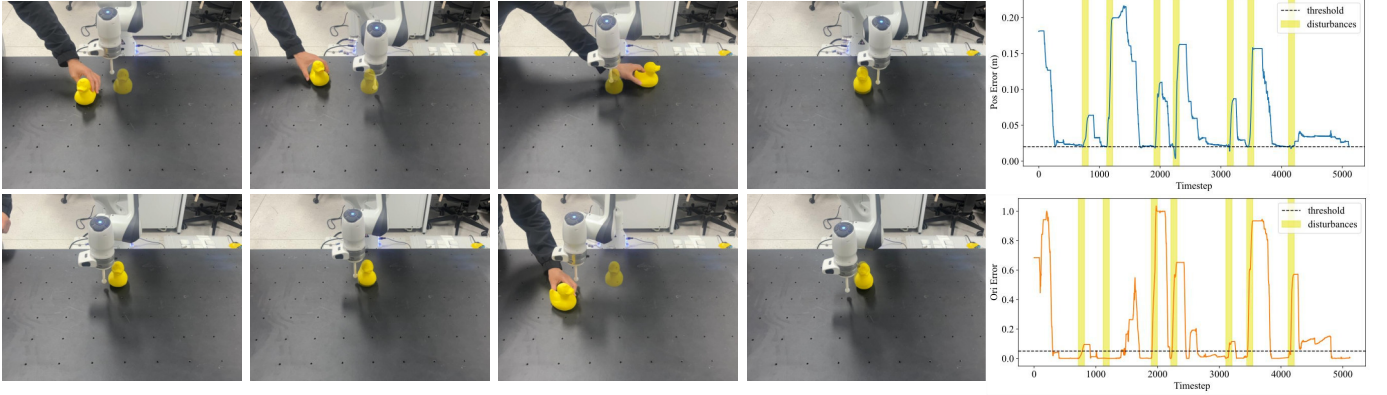


Fig. 13. Snapshots of manipulation under external disturbances. The object’s pose is continuously perturbed during manipulation to demonstrate SCSP’s real-time robustness under such disturbances. The panels on the right shows the temporal evolution of position and orientation errors.

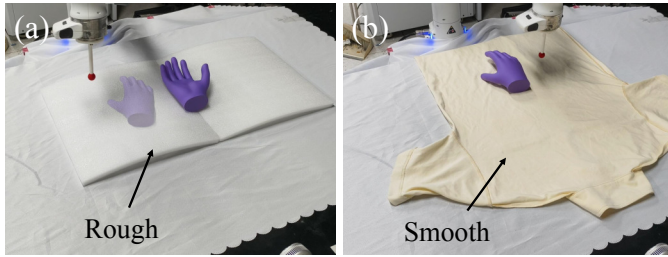


Fig. 14. Robustness validation of SCSP on surfaces with different friction. (a) Foam. (b) Cotton clothes.

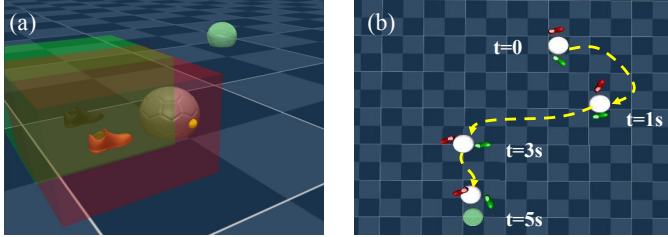


Fig. 15. Soccer dribbling task. (a) Experiment setup. The task is to kick the soccer to the target (translucent). Red and green cubes show footworkspaces, and the yellow point indicates CSO-planned contact locations. (b) Dribbling trajectory.

separately for the left and right feet: using the midpoint between both feet as the center, the space is divided into left and right half-spaces. When x_k appears in a half-space, the corresponding left or right foot is activated. Additionally, workspace constraints (represented as semi-transparent cubes) are imposed on each foot to prevent singular gait patterns. The number of CSO sampling points is set to $n_s = 70$, and points close to the ground $\{p_i \in \mathcal{I}_{obj} \mid z_i < 0.02m\}$ are masked out. We randomly selected 10 goal positions for the soccer ball, and SCSP continuously controlled the robot’s feet to dribble the ball to each target, achieving a 100% success rate. The experimental results are shown in Fig. 15. This experiment serves as a toy example, simulating the approaching and contact phases in humanoid soccer dribbling. In the future, we plan to extend SCSP to more complex motion planners to realize a more complete demonstration [50], [51].

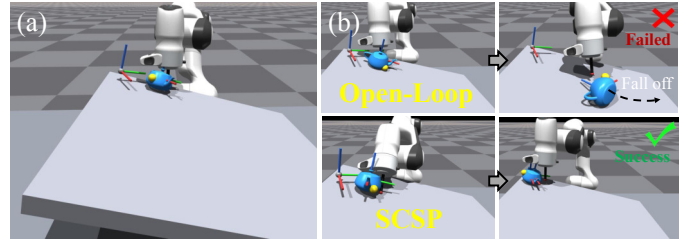


Fig. 16. Experiment setup. The task is to push an object on a surface tilted at 15° . (a) The open-loop method fails when execution errors occur, causing the teapot to fall off the tilted surface. (b) SCSP can replan online and thus successfully complete the dynamic task.

2) *Tilted Pushing*: SCSP can be extended to manipulation tasks in uneven environments, such as tilted pushing. Stable object poses on a slope are inherently fragile, requiring accurate contact location selection and online replanning. We compare open-loop methods, which plan a contact sequence offline, with SCSP, which replans online during execution. As the baseline, we use STOCS-3D to compute an offline sequence of contact locations and contact forces, and execute this sequence using our CPO. Experiments are conducted in IsaacGym with a 15° slope and a friction coefficient of 0.1. For SCSP, we set $w_{stab} = 0$ and $\hat{\lambda}_{max}^n = 0.5$, while keeping all other parameters the same as in Section VI-B. We selected the teapot and duck (used in previous experiments) and randomly set 10 stable poses on the slope as goal poses, and pushed and rotated the objects from below to reach these goals. The results in Table VII and Fig. 16 show that the real-time replanning capability of SCSP provides substantially better robustness.

3) *Grasping & Dexterous Manipulation*: SCSP can also be applied to predict grasping poses, such as bimanual grasping and multi-finger grasping. Following [52], we use the Grasp Wrench Boundary (GWB) [53] to select grasping regions and identify the top-K pairs of potential valid regions for grasping. Subsequently, sampling of the candidate set $\mathcal{I}_{obj}$ is performed within these regions. The objective of the CSO inner loop

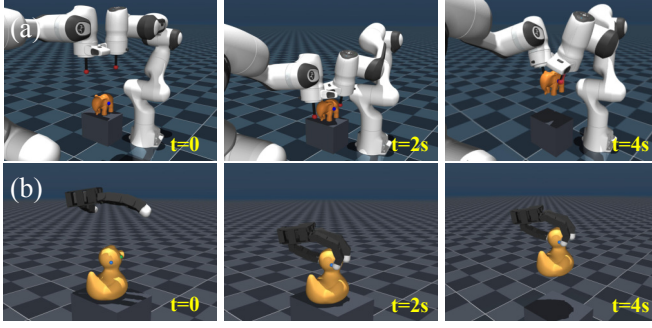


Fig. 17. Snapshots of the multi-contact grasping tasks. (a) Bi-manual grasping of an elephant with the size scaling factor $\varsigma = 1$. (b) Multi-finger grasping of a duck with $\varsigma = 1.5$.

optimization is then transformed to force closure:

$$\begin{aligned} \min_{\hat{\lambda}_r \in \mathcal{K}} \quad & \sum_{j=1}^{N_d} \left\| \xi \mathbf{w}^{(j)} - \mathbf{J}_o \hat{\lambda}_r^{(j)} \right\|_2^2 + w_\lambda \sum_{j=1}^{n_d} \left\| \hat{\lambda}_r^{(j)} \right\|_2^2 \\ \text{s.t.} \quad & \sum_{i=1}^K \hat{\lambda}_r^{(j),n} \geq \zeta, \quad \forall j = 1, \dots, n_d. \end{aligned} \quad (52)$$

where n_c is the number of robot contact points, n_d denotes the number of sampled disturbance wrenches, $\mathbf{w}^{(j)} \in \mathbb{R}^6$ represents the j -th disturbance wrench. $\lambda_r^{(j)}$ denotes the robot contact force corresponding to disturbance $\mathbf{w}^{(j)}$. ξ is the disturbance scaling factor, ζ the minimum total normal force required to maintain an active grasp and $w_\lambda = 0.1$. During each rollout, the CSO in SCSP solves (52) for the nearest points $\mathbf{p}_{\text{near}}^i$ to the robot end-effector or Allegro fingers additional to the top 5 pairs of potential grasping pairs, thereby searching for grasping poses online. We use an open-loop baseline that computes the optimal grasping locations offline using CSO and executes them online with CPO for comparison. For bi-manual and multi-finger grasping, we selected five objects each, placed them on a pedestal (black), randomly perturbed their yaw angles θ_{yaw} and scaled their sizes by a random factor $\varsigma \in \{0.5, 1, 1.5, 2\}$, conducting 20 trials for each task. The results are shown in Table VII and Fig. 17. SCSP achieves a higher success rate than the baseline in these trials; however, it fails on some objects with complex geometry because automatic convexification of the collision mesh by MuJoCo.

B. Case Study II: Articulated Object Manipulation

SCSP uses active contact selection to perform articulated object manipulation. We evaluate this capability on a drawer and a microwave door, which require sequential contact interactions. The tasks are to pull the drawer to a target displacement and to open the microwave door to a target angle. In both tasks, the object is semantically segmented into different parts, such as handles, while non-manipulable parts are excluded. For the drawer, we directly compute the final pulled-out pose as the goal pose. For the microwave, which involves rotational motion, we set the goal pose of each rollout

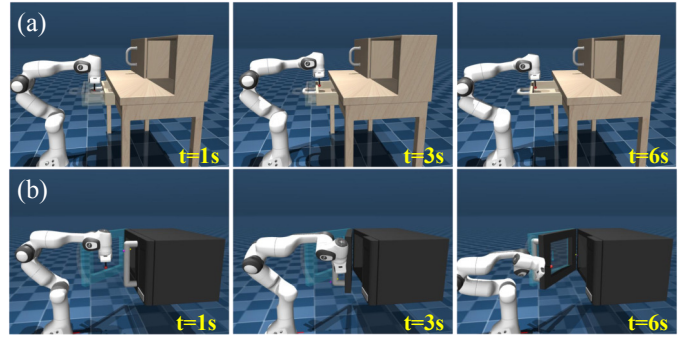


Fig. 18. Snapshots of long-horizon tasks. Given only the target pose of the object, SCSP performs online reasoning and selects appropriate contact locations to accomplish the task. (a) Opening a drawer. (b) Opening the microwave door.

as the SLERP interpolation between the current state and the desired door state. The parameter and cost settings of SCSP are the same as those in Section VI-B, with $w_{\text{stab}} = 0$. The solver for CPO is an MPPI-based motion planner [54]. As a baseline, we use an open-loop method that plans the contact sequence offline using CSO and executes it online with CPO, in order to demonstrate the effectiveness of online contact selection. We randomly perturbed the distance and yaw angle between the drawer or microwave and the Franka base by $\Delta d \in [-0.1, 0.1]$ m and $\Delta \theta_{\text{yaw}} \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ and conducted 10 trials for each object. The experimental results are shown in Fig. 18, Table VII and the website videos. Although the success rates are comparable, the open-loop method cannot handle certain execution errors, such as contact slippage or the robot approaching singular configurations.

VIII. DISCUSSION & LIMITATIONS

The theoretical and experimental results support the usefulness of explicit contact selection, while also exposing several limitations of the current implementation.

CSO provides contact-location priors through mesh discretization and a lightweight surrogate contact model. This design is computationally efficient, but its accuracy decreases in complex contact configurations because it assumes a fixed object–environment contact set, neglects inter-contact coupling, and represents contacts as points. CSO therefore does not provide sequential, high-fidelity state prediction and is not directly suited to complex end-effectors or deformable objects. A higher-fidelity contact model, such as the model in [55], could be used when these capabilities are required.

CPO generates trajectories for contact-free and contact-rich stages using model-predictive control. Its current implementation may be less suitable for tasks such as locomanipulation and may not maintain continuous surface contact during contact switching. Future work can replace the rule-based ranking strategy with a learned representation and can combine CPO with a more expressive motion planner such as [56].

The experiments also suggest that the interface between contact selection and trajectory optimization is important: CPO can use a feasible nearby contact point when the lowest-cost

TABLE VII
SUCCESS TRIALS ACROSS DIFFERENT TASKS.

Metric	Soccer Dribbling		Tilted Pushing		Bi-Manual Grasping		Multi-finger Grasping		Long-horizon Tasks	
	Open-Loop	SCSP	Open-Loop	SCSP	Open-Loop	SCSP	Open-Loop	SCSP	Open-Loop	SCSP
Success Trials	/	10/10	0/20	18/20	0/20	15/20	15/20	16/20	15/20	20/20

sampled point is difficult to execute. This observation does not establish that the surrogate model is accurate outside the evaluated tasks, and the conclusions about robustness should therefore be interpreted within the tested range of geometries, disturbances, and model mismatch.

IX. CONCLUSION

This work presented SCSP, a cascaded framework that separates contact-location selection from contact-trajectory optimization for contact-rich manipulation. CSO evaluates a finite set of sampled surface points with a surrogate contact model, and CPO uses the resulting contact prior together with a ranking rule to generate approaching, contact, and contact-switching trajectories. Simulations and hardware experiments show that this decomposition can improve manipulation diversity and task success under the tested geometric and modeling conditions. The results also indicate that SCSP remains limited by the point-contact and fixed-contact-set assumptions of the surrogate model, as well as by the need for a one-time manual MuJoCo calibration in the vision-based hardware experiments.

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APPENDIX

A. Proof of Closest Approximation

Let $\mathbf{W}_{ii} \in \mathbb{R}^{n \times n}$ be the original Delassus matrix. We seek a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ that minimizes the Frobenius norm of the approximation error:

$$D^* = \arg \min_D \|\mathbf{W}_{ii} - D\|_F^2. \quad (53)$$

Expanding the Frobenius norm, we have

$$\|\mathbf{W}_{ii} - D\|_F^2 = \sum_{j \neq k} |(\mathbf{W}_{ii})_{jk}|^2 + \sum_{j=1}^n |(\mathbf{W}_{ii})_{jj} - d_j|^2. \quad (54)$$

The off-diagonal terms are not affected by choosing D , and so the minimization reduces to independent scalar problems for each diagonal entry:

$$d_j^* = \arg \min_{d_j} |(\mathbf{W}_{ii})_{jj} - d_j|^2, \quad j = 1, \dots, n. \quad (55)$$

It is immediate that the minimizer is $d_j^* = (\mathbf{W}_{ii})_{jj}$. Hence (27) is the closest diagonal approximation in the Frobenius norm with a small regularization term for numerical stability.

B. Local Motion-Prediction Accuracy of SCM

We compare (29) and (31) using identical current states, robot contact inputs, contact geometry, and model parameters. Let λ_{env} and $\hat{\lambda}_{\text{env}}$ denote the nonnegative multipliers in (21) and their SCM predictions, respectively. Let $D \succ 0$ collect the diagonal approximations in (27), $E := W_{\text{env}} - D$, and $M_o \succ 0$, $R \succ 0$. We use $\|z\|_A = (z^\top A z)^{1/2}$ and define the geometry-dependent motion gain

$$\Gamma_{\text{env}} := \|M_o^{-1/2} \tilde{J}_{\text{env}}^\top D^{-1/2}\|_2. \quad (56)$$

1) *Directional accuracy*: In the zero-gap setting of (28), reference complementarity and (29)–(31) give

$$\begin{aligned} h^2(\mathbf{v}^+)^\top M_o(\hat{\mathbf{v}}^+ - \mathbf{v}^+) &= (\hat{\lambda}_{\text{env}} - \lambda_{\text{env}})^\top h \tilde{J}_{\text{env}} \mathbf{v}^+ \\ &\geq (\lambda_{\text{env}} - \hat{\lambda}_{\text{env}})^\top R \lambda_{\text{env}} \\ &\geq -\frac{1}{4} \hat{\lambda}_{\text{env}}^\top R \hat{\lambda}_{\text{env}}, \end{aligned} \quad (57)$$

where the inequalities follow from nonnegative multipliers and slack, followed by completing the square. For nonzero velocity changes, their inertia-weighted angle satisfies

$$\begin{aligned} \cos \theta_{M_o} &:= \frac{(\mathbf{v}^+)^\top M_o \hat{\mathbf{v}}^+}{\|\mathbf{v}^+\|_{M_o} \|\hat{\mathbf{v}}^+\|_{M_o}} \\ &\geq \frac{\|\mathbf{v}^+\|_{M_o}}{\|\hat{\mathbf{v}}^+\|_{M_o}} - \frac{\hat{\lambda}_{\text{env}}^\top R \hat{\lambda}_{\text{env}}}{4h^2 \|\mathbf{v}^+\|_{M_o} \|\hat{\mathbf{v}}^+\|_{M_o}}. \end{aligned} \quad (58)$$

For the stationary-environment tasks, the force limits in (32) constrain robot-induced loading. With bounded non-contact loads, fixed contact geometry, and retained diagonal responses bounded away from zero, (26) also bounds the predicted environment reactions. Small R then limits the quadratic regularization contribution. Its normalized contribution in (58) is small during motion-producing steps for which the velocity changes are not negligible relative to the contact loading.

When the normalized correction is negligible, a large angular error requires substantial overestimation of the motion magnitude. Conversely, closely matched magnitudes imply small angular errors. The weak-coupling analysis below supports such magnitude agreement away from near-static force balance.

Most tasks considered here emphasize pose regulation rather than highly dynamic motion. Accordingly, the stable-pose objective in (34) uses the predicted motion to identify contacts that promote goal-directed pose changes, rather than to track an exact velocity. Preserving these motion trends is therefore particularly important for CSO contact selection.

2) *Magnitude accuracy*: Since $W_{\text{env}} = D + E$, the reference complementarity conditions and SCM prediction can be written as

$$\begin{aligned}\lambda_{\text{env}} &= \max[-D^{-1}(\mathbf{g}_{\text{env}} + E\lambda_{\text{env}}), 0], \\ \hat{\lambda}_{\text{env}} &= \max(-D^{-1}\mathbf{g}_{\text{env}}, 0).\end{aligned}\quad (59)$$

The scalar map $\max(\cdot, 0)$ does not amplify differences. Applying this property componentwise and summing with weights D_{ii} gives

$$\|\hat{\lambda}_{\text{env}} - \lambda_{\text{env}}\|_D \leq \|D^{-1/2}E\lambda_{\text{env}}\|_2. \quad (60)$$

Together with (29) and (31), this implies

$$h\|\hat{\mathbf{v}}^+ - \mathbf{v}^+\|_{M_o} \leq \Gamma_{\text{env}}\|D^{-1/2}E\lambda_{\text{env}}\|_2. \quad (61)$$

The discarded coupling and diagonal-perturbation effects are therefore weighted by the current contact loading. Small $\|D^{-1/2}E\lambda_{\text{env}}\|_2$ yields small force errors and, with a moderate Γ_{env} , small motion errors. This characterizes the weak-coupling regime targeted by Assumption 2 and (25).

Contact orientations, moment arms, and object inertia determine E , D , and Γ_{env} , affecting both coupling strength and the conversion of force errors into motion errors. Robot-force magnitude affects λ_{env} through $\tilde{\mathbf{b}}_o$ in (19), with the robot-induced load limited by (32). Prediction accuracy therefore depends jointly on contact geometry and current loading, including gravity, rather than on coupling strength alone.

3) *Contact transitions*: A newly established object-environment contact introduces a reaction absent from the SCM rollout. Its effect is not included in $E\lambda_{\text{env}}$ for the previous contact set, and the fixed-set estimates do not control the full transition error. SCSP incorporates updated object states and contact estimates through high-frequency replanning, while the ranking strategy in (40) temporally filters contact-switching decisions. Together with CPO using $\hat{\mathbf{p}}_r^*$ as guidance rather than a prescribed contact, these mechanisms limit the persistence of inaccurate priors and reduce unnecessary switching. When contact estimates are refreshed promptly relative to object motion, they help prevent transient SCM errors from persistently biasing contact selection.

C. Proof of Lemma 4

Consider the lift-and-place cost $\ell_{\text{lp}} = (1 - \gamma)\ell_{\text{lift}} + \gamma\ell_{\text{place}}$. We analyze the different cases of the switching parameter γ .

Case 1: When $\gamma = 1$, $\ell_{\text{lp}} = \ell_{\text{place}}$. The placing objective only guides the robot to reference points $\mathbf{p}_{\text{ref}} \in \{\hat{\mathbf{p}}_r^*, \mathbf{p}_{\text{near}}\}$ on the object, it does not obstruct the first-order improvement of

the objective since $\hat{\mathbf{p}}_r^*$ and $\mathbf{p}_{\text{near}}$ is the optimal or an acceptable suboptimal solution returned by CSO for improving ℓ_{obj} .

Case 2: When $\gamma = 0$, $\ell_{\text{lp}} = \ell_{\text{lift}}$ and the lifting phase moves the end-effector away from the object to an intermediate pose above the reference contact. Since CPO uses a short horizon, it is usually unable to reach $\mathcal{S}_{\text{contact}}$ while moving away from the object, and thus unable to obtain the gradient of ℓ_{obj} . Therefore, ℓ_{obj} does not affect ℓ_{lp} .

Since ℓ_{lp} is a convex combination of lift and place costs, ℓ_{lp} and ℓ_{obj} are non-conflicting objectives.